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A PROJECT REPORT

On

"Trust Chain: Blockchain Powered E-Commerce Reputation System"

Submitted in Partial Fulfillment for the Award of the Degree

of

BACHELOR OF ENGINEERING

IN

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Certificate

Certified that the Project Work entitled **"Trust Chain: Blockchain Powered E-Commerce Reputation System"** carried out by **ADARSH M (1SG21CS002), PRINCY DEY (1SG21CS067), SHASHANK U (1SG21CS090), SINCHANA NAGESH (1SG21CS102),** bonafide students of **Sapthagiri College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering** in **Computer Science and Engineering** of **Visvesvaraya Technological University**, **Belagavi** during the academic year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the Department Library. The project report has been approved as it satisfies the academic requirements in respect of **Project Work (21CSP76)** prescribed for the said degree.

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We consider it is a privilege to express our gratitude and respect to all those who guided us in completion of Final Project.

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ABSTRACT

Reviews and reputation scores of sellers play an important role in decision-making process of potential buyers in an e-commerce system. A trustworthy and reliable reputation system is a crucial component in the e-commerce ecosystem, as buyers rely on it to make informed decisions. In this work, we develop a privacy-preserving decentralized reputation system designed to include countermeasures against some known attacks. Our model is built on two permissioned blockchains, one for maintaining feedback tokens which are generated when customer purchases a product and other for saving discount tokens which will be generated when customer gives feedback of product. One of the key advantages of the proposed approach is the use of verifiable credentials for digital identities of sellers, feedback tokens issued to buyers after performing an e-commerce transaction and discount tokens issued to buyers after feedback submission. This helps to ensure that the feedback and identity information is authentic and tamper-proof, reducing the likelihood of identity-related attacks. Additionally, the collection of feedbacks and application of business rules are implemented as smart contracts in blockchain. This provides a secure and transparent mechanism for processing feedback, reducing the likelihood of unfair feedbacks. Overall, the proposed approach presents a robust reputation system that can help reduce identity-related attacks and unfair feedbacks. The privacy-preserving nature of the system ensures that sensitive information is protected while still enabling the verification of digital identities. The use of feedback and discount tokens incentivizes buyers to provide accurate and honest feedback, which can help reduce unfair feedbacks and identity-related attacks. Finally, the use of smart contracts ensures transparency and immutability, which enhances the overall reliability of the system.

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DECLARATION

We **ADARSH M (1SG21CS002), PRINCY DEY (1SG21CS067), SHASHANK U (1SG21CS090)** and **SINCHANA NAGESH (1SG21CS102),** bonafide students of **Sapthagiri College of Engineering,** hereby declare that the project entitled "**Trust Chain: Blockchain Powered E-Commerce Reputation System**" submitted in partial fulfilment for the award of Bachelor of Engineering in **Computer Science & Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2024-2025 is our original work and the project has not formed the basis for the award of any other degree, fellowship or any other similar titles.

Name of the Student	Signature of the Student with date
1) ADARSH M (1SG21CS002)	______________________________
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INTRODUCTION

The rise of e-commerce has significantly transformed how buyers interact with sellers. In this digital environment, where direct human interaction is minimal, **trust** becomes a crucial element. Traditionally, reputation systems based on reviews and ratings have served as mechanisms for establishing this trust. However, these systems are vulnerable to manipulation and lack transparency. **Trust chain** introduces a decentralized, privacy-preserving reputation system underpinned by **blockchain technology** and **verifiable credentials**, offering a tamper-proof and transparent environment for trust management in e-commerce.

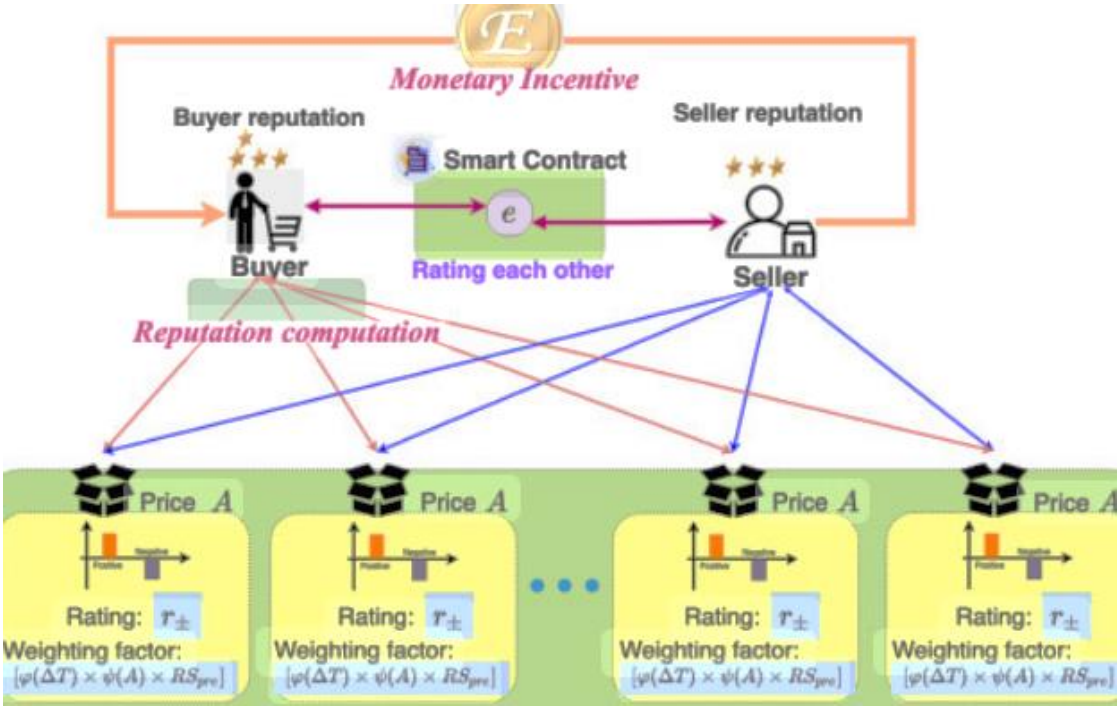

Figure 1.1 – Blockchain based reputation system

1.1 CENTRALIZED REPUTATION SYSTEMS IN E-COMMERCE

Centralized e-commerce systems are controlled by a single authority, typically the platform provider. These platforms, such as Amazon, eBay, and Flipkart, manage all core services—product listings, transactions, payment processing, user identity, and most critically, the reputation system based on feedback and ratings. Seller reputation in these systems is determined by reviews submitted by buyers, stored in centralized databases and processed by algorithms defined solely by the platform.

This approach, while functional, creates a strong dependency on the platform's integrity, transparency, and resilience. The users (buyers and sellers alike) place their trust in the platform to accurately and fairly manage reputational data. However, centralized control also opens the door to numerous risks and inefficiencies, particularly in contexts where trust and impartiality are paramount.

1.1.1 Advantages of Centralized Systems

- **Ease of Development and Maintenance**: Centralized systems are straightforward to design and deploy since control is streamlined.
- **Efficient Moderation**: Platform administrators can quickly detect and remove inappropriate content, such as spam or offensive reviews.
- **Unified Data Access**: Centralized repositories allow for rapid querying and reporting of data, aiding analytics and business intelligence.
- **Controlled User Experience**: Platforms can maintain consistency in UI/UX and enforce uniform design standards and usage policies.
- **Integrated Customer Support**: Platforms can directly respond to disputes, refund requests, and manage review resolutions centrally.
- **Rapid Updates**: Any system-wide changes or policy updates can be deployed immediately from a single point of control.

1.1.2 Disadvantages of Centralized Systems

- **Single Point of Failure**: A crash, outage, or breach in the central server can render the entire system inoperable or lead to massive data losses and compromised services.
- **Bias and Censorship**: Platforms have the authority to suppress or manipulate reviews based on internal policies or external pressures, which may compromise neutrality and fairness.
- **Fraudulent Reviews**: Without effective identity verification mechanisms, sellers can boost their ratings using fake accounts, bots, or paid reviewers (ballot-stuffing).
- **Loss of User Control**: Users do not have ownership over their own data or feedback and are entirely dependent on the platform's terms and access control.
- **Vendor Lock-in**: Seller reputations are not portable. Even a top-rated seller on one platform must start from scratch on another, leading to fragmentation of trust.
- **Opaque Algorithms**: Reputation scores are often calculated using proprietary algorithms that are not disclosed to the public, making the scoring process opaque and potentially biased.

1.1.3 Security and Privacy Concerns

- **Data Breaches**: Central databases are attractive targets for cyberattacks that can expose sensitive buyer and seller information, including personal data and transaction history.
- **Sybil Attacks**: Malicious users may create multiple fake identities to manipulate the feedback system or to influence ratings and product perception unfairly.
- **Privacy Violations**: Users' data may be tracked, sold, or shared with third parties without explicit consent, raising significant ethical and regulatory concerns.
- **Unverifiable Feedback Origins**: There's often no way to confirm whether a reviewer has actually purchased a product or interacted with a seller, reducing the authenticity of reviews.
- **Reputation Re-entry Loopholes**: Sellers who are banned for malpractice or poor service can often rejoin the platform under a new identity, escaping accountability and continuing harmful practices.

Moreover, the centralization of power creates a monopoly over data and trust, undermining the open, community-driven ethos of digital marketplaces. Trust, once lost due to biased moderation or manipulated reputations, is hard to rebuild. This makes it evident that a more robust, transparent, and user-controlled system is essential for the future of digital commerce.

These drawbacks necessitate the transition to a new model of managing trust and reputation—one that eliminates reliance on a single authority, ensures data integrity, and empowers users with greater control and transparency. The answer lies in decentralization, enabled by blockchain and Web 3.0 technologies.

1.2 THE NEED FOR DECENTRALIZED APPLICATIONS

As e-commerce continues to expand globally, the limitations of centralized platforms have become more apparent. Decentralized applications offer a compelling alternative by removing single points of control and creating systems that are transparent, trustless, and resilient. The essence of a decentralized application is that it operates on a distributed network, typically a blockchain, which enables peer-to-peer interactions without reliance on centralized authority.

1.2.1 Key Motivations

- **Trust Independence**: dApps are built on protocols that use consensus mechanisms like Proof of Stake or Practical Byzantine Fault Tolerance to validate transactions, eliminating reliance on a central authority.
- **User Empowerment**: With dApps, individuals have full ownership over their data, identities, and transactions. Reputation data is cryptographically secured and publicly verifiable.
- **Data Integrity**: Blockchains ensure that once data is added, it cannot be altered or removed without consensus, making them highly tamper-resistant.
- **Attack Resilience**: The decentralized nature of these applications makes them inherently more resistant to Distributed Denial of Service (DDoS) attacks and other cybersecurity threats.

1.2.2 Implications for E-Commerce

- **Genuine Reviews**: Feedback can be tied to verified transactions via smart contracts and token-based systems.
- **Transparency**: Review history, seller performance, and transaction data are open and accessible, reducing information asymmetry.
- **Cross-Platform Interoperability**: Sellers can carry their reputations across multiple marketplaces, reducing vendor lock-in and creating a unified trust layer.
- **Incentivized Participation**: Users are rewarded with tokens for honest participation, improving engagement and the quality of feedback.

These characteristics make decentralized applications a powerful tool for transforming trust management in e-commerce, shifting control from centralized entities to the community of users.

1.3 WEB 3.0 THE DECENTRALIZED INTERNET

Web 3.0 represents the evolution of the internet from static pages (Web 1.0) and interactive platforms (Web 2.0) to a trustless, user-centric digital space. This phase emphasizes decentralization, digital ownership, and programmable trust through technologies like blockchain.

1.3.1 Potential Applications

- Decentralized social networks

- Tokenized loyalty and rewards
- Self-sovereign identity systems
- E-commerce platforms with smart contract-based automation

1.3.2 Blockchain Technology

Blockchain is the foundational layer of Web 3.0, offering:

- Distributed ledger systems
- Consensus protocols (e.g., Proof of Stake, Practical Byzantine Fault Tolerance)
- Smart contracts for automated rule enforcement

1.3.3 Applications of Blockchain

- Cryptocurrency and decentralized finance (DeFi)
- Digital voting systems
- Healthcare data management
- E-commerce reputation and trust frameworks (e.g., Trustchain)

1.4 BACKGROUND

Traditional e-commerce platforms have long struggled with ensuring trust among unknown buyers and sellers. Reputation scores, while helpful, are easy to manipulate. Numerous studies and real-world scenarios highlight the vulnerabilities of centralized systems to fraud, censorship, and privacy breaches.

Verifiable credentials and blockchain present a transformative solution. By embedding identities and feedback in immutable, verifiable structures, these tools provide a foundation for secure and scalable trust systems.

1.5 OVERVIEW OF THE PRESENT WORK

- This project introduces **Trust chain** – a decentralized reputation system for e-commerce, leveraging:
- **Two permissioned blockchains**: one for feedback token issuance, another for

reward tokens

- **Verifiable credentials**: for buyer and seller identity validation
- **Smart contracts**: to automate feedback validation and reputation updates
- **Token incentives**: to motivate honest feedback

The system is designed to be privacy-preserving, secure, scalable, and resistant to manipulation.

1.6 PROBLEM STATEMENT

Current centralized e-commerce platforms fail to offer transparent, tamper-proof, and identity-secure reputation systems. Fake reviews, biased moderation, and re-entry of bad actors through new identities diminish trust. There is a critical need for a decentralized, verifiable, and incentive-aligned reputation model.

1.7 OBJECTIVES

- To design a decentralized architecture for managing buyer feedback and seller reputation.
- To use blockchain-based verifiable credentials for ensuring authenticity of participants.
- To issue feedback tokens only to verified buyers, ensuring review legitimacy.
- To incentivize feedback through reward tokens accepted across participating platforms.
- To eliminate identity-based attacks like Sybil and ballot-stuffing.
- To create tamper-proof, transparent reputation trails visible to all stakeholders.
- To ensure cross-platform compatibility for seller reputations.
- To provide a scalable, modular, and privacy-respecting solution for modern e-commerce.

1.8 Organization of the Report

- **Chapter 1:** Introduction to the problem domain, objectives, and background.
- **Chapter 2:** Literature review of existing systems and gaps identified.
- **Chapter 3:** System requirements and technology stack.

- **Chapter 4:** System architecture and design methodology.
- **Chapter 5:** Implementation details of various modules.
- **Chapter 6:** Testing and validation of system components.
- **Chapter 7:** Results, conclusion, and future enhancements.
- **References**: Citations and research sources

CHAPTER 2

LITERATURE REVIEW

CHAPTER 2

LITERATURE REVIEW

2.1 Summary of Prior Works

A variety of blockchain-based reputation systems have been proposed to address trust issues in e-commerce platforms. Key contributions include:

[1] Sun et al.: Feedback Token Model for Verified Reviews

Sun et al. proposed a novel feedback token model that aimed to counteract common e-commerce trust issues, particularly ballot-stuffing, where sellers create fake buyer accounts to inflate product ratings. Their system introduced the concept of only allowing verified buyers—those who have completed a genuine transaction—to leave reviews. These buyers received a special feedback token that could be redeemed to submit a rating or comment, ensuring only legitimate participants could influence the seller's reputation. The implementation was carried out on public blockchain platforms, leveraging their transparency and immutability. However, a significant drawback emerged: high transaction fees. Each review submitted and token redeemed incurred a gas cost, which discouraged participation and made the system economically unsustainable at scale. Moreover, scalability became a concern, as processing a large number of transactions in real time strained the blockchain infrastructure.

[2] Li et al. (RepChain): Privacy-Preserving Reputation via Blind Signatures

RepChain, developed by Li et al., addressed privacy concerns in blockchain-based reputation systems by implementing a privacy-preserving model using blind signatures. This cryptographic method allowed users to anonymously submit reviews, thereby protecting their identities while maintaining the integrity of their feedback. The approach was significant in contexts where buyers may fear retaliation or value confidentiality. Despite its innovation in anonymity, RepChain lacked a vital component—an incentive mechanism. There were no meaningful rewards or penalties to encourage honest reviews or deter malicious actors. This limitation reduced the practical effectiveness of the model, as users had little motivation to provide truthful, timely feedback.

[3] Dhakal et al. (DTrust): Decentralized Trust with Ethereum Smart Contracts

Dhakal et al. introduced DTrust, a decentralized trust framework leveraging Ethereum smart contracts to automate the validation of product reviews. DTrust's key contribution was its use of programmable logic to assess review legitimacy based on transaction data and buyer credentials, eliminating the need for centralized moderation. This enhanced transparency and reduced the chance of fraudulent reviews. However, the architecture depended heavily on on-chain operations, meaning all data validation and logic execution occurred within the blockchain. This resulted in high operational costs due to Ethereum's gas fees and made the system difficult to scale. As the number of reviews grew, the network became increasingly congested, limiting its ability to support large e-commerce ecosystems efficiently.

[4] Zulfiqar et al. (EthReview): Extending DTrust with Review-Oriented Smart Contracts

Building upon the foundations laid by DTrust, Zulfiqar et al. introduced EthReview, which specialized in managing product reviews via Ethereum smart contracts. The platform aimed to offer a decentralized and immutable review ledger that could be integrated into e-commerce platforms for transparent feedback. While it successfully enabled the automated handling of reviews, the system did not sufficiently address reviewer privacy. Feedback was permanently recorded on-chain without sufficient anonymization techniques, which discouraged open and honest reviews, especially in controversial or competitive product domains. This lack of privacy undermined user trust and potentially led to biased or withheld feedback.

[5] Kugblenu and Vuorimaa: Governance via Permissioned Blockchains

Kugblenu and Vuorimaa explored the use of permissioned blockchains for managing reputation in e-commerce, diverging from the more commonly used public blockchain models. In a permissioned setting, only approved nodes can participate in the network, which improved governance, reduced energy consumption, and eliminated transaction fees associated with public chains. Their approach helped address performance and compliance issues and made the system more suitable for enterprise use cases. However, the system struggled with ensuring the authenticity of user feedback. Without employing robust cryptographic methods, such as zero-knowledge proofs or digital signatures, the platform could not guarantee that reviews were genuine, leaving it vulnerable to manipulation.

[6] Beaver: A Fully Decentralized Marketplace with Built-in Reputation System

Beaver proposed a comprehensive, fully decentralized e-commerce platform that embedded a reputation system as a core feature. Unlike add-on systems, Beaver's model was integrated into every layer of the marketplace, from user identity management to transactional verification and feedback recording. This holistic approach ensured seamless interaction between buyers and sellers while leveraging blockchain's benefits. However, one of its main limitations was its independence—it required building an entirely new marketplace from the ground up, rather than integrating with existing platforms like Amazon or eBay. This drastically limited its adoption potential, as users and developers had to migrate to or recreate existing ecosystems within Beaver's framework.

[7] Schaub et al. and Owiyo et al.: Blinded Token Systems for Anonymous Feedback

Schaub et al. and Owiyo et al. introduced blinded token systems designed to enable anonymous feedback mechanisms in blockchain-based e-commerce. By issuing tokens that could be “blinded” (i.e., used without revealing the identity of the holder), these systems preserved user privacy and encouraged more honest feedback. This innovation helped address one of the major drawbacks of on-chain review systems—traceability of user identities. Despite this advantage, the proposals lacked well-defined token issuance mechanisms. There was little clarity on how users earned these tokens or what incentives existed for continued participation. As a result, the systems raised questions about their sustainability and long-term user engagement.

[8] Carboni and Ahn et al.: Bitcoin-Based Reputation with Pseudonymous Identities

Carboni and Ahn et al. focused on leveraging the Bitcoin blockchain to build reputation systems around pseudonymous identities. Users interacted through Bitcoin addresses, and reputational scores were derived from their transaction history and peer interactions. This model preserved a level of anonymity while maintaining verifiability. However, the lack of programmable logic (e.g., smart contracts) in the Bitcoin blockchain limited system functionality. Reputation tracking was static and inflexible, and most implementations relied on cryptocurrency-based payments to signal trust, which didn’t map well onto nuanced user behavior or review content. The lack of automation also posed barriers to integration with dynamic e-commerce environments.

[9] Ramachandiran: Dual Blockchain Model for Separating Identity and Transactions

Ramachandiran proposed a dual blockchain model that separated user identity data from transaction data to improve privacy and modularity. One blockchain stored pseudonymous identity credentials, while the other handled transaction details and reviews. This separation aimed to reduce the risk of linking personal identity with consumer behavior, thereby enhancing user privacy. However, the model introduced complexities in system synchronization and redundant data storage. Each interaction required cross-chain communication and validation, which increased latency and infrastructure costs. Additionally, privacy enhancements were insufficient—metadata and linking patterns still posed de-anonymization risks if not properly mitigated.

[10] Park et al., Nguyen et al., Omori & Kishigami: Cross-Domain Blockchain Trust Applications

The work of Park et al., Nguyen et al., and Omori & Kishigami extended the concept of decentralized reputation beyond traditional e-commerce into domains such as the Internet of Things (IoT), auction systems, and social data marketplaces. These applications highlighted the broader relevance of blockchain-powered trust mechanisms across industries. For instance, in IoT networks, reputation systems were used to assess the trustworthiness of devices, while in data markets, reputation scores helped determine data seller credibility. These cross-domain adaptations underscored the flexibility and appeal of decentralized trust. However, many of these systems faced challenges similar to those in e-commerce—namely, high costs, privacy risks, and a lack of unified standards for reputation computation and verification.

2.2 Outcome of the review

Based on the analysis of the above studies, several insights and limitations are identified:

Strengths Identified in Prior Work

- Blockchain enhances **immutability**, **auditability**, and **transparency**.
- Smart contracts enable **automated feedback processing**.
- Blind signatures and pseudonyms provide a degree of **privacy preservation**.
- Permissioned blockchains improve **performance** and **governance**.

Limitations Observed

- **High gas costs** on public blockchains reduce feasibility for large-scale deployment.
- Many systems **lack incentive mechanisms**, limiting honest user participation.
- **Privacy is often overlooked**, leading to potential retaliation or biased reviews.
- **Integration with existing platforms** is minimal, requiring full infrastructure replacement.
- **Feedback authenticity** is rarely enforced through cryptographic proofs.

- Most models fail to provide **cross-platform reputation portability**.
- Reviewer **identity verification** is either absent or inadequately implemented.

These outcomes reveal a critical gap: the need for a **scalable, privacy-preserving, incentive-compatible reputation system** that can be integrated into **existing e-commerce infrastructures**.

2.3 Proposed work

To address the gaps in prior research, the proposed system—**Trust chain**—introduces a decentralized reputation model with the following distinguishing features:

1. **Dual Permissioned Blockchains**
   - One blockchain issues **feedback tokens** upon successful purchase.
   - A second blockchain manages **discount tokens** as reviewer incentives.
2. **Verifiable Credentials for Digital Identities**
   - Both buyers and sellers are assigned cryptographically verifiable identities.
   - Based on W3C standards for structured, private, and tamper-resistant credentials.
3. **Smart Contract-Based Feedback Validation**
   - Feedback submission is permitted only upon presentation of a valid **feedback token**.
   - Smart contracts automate scoring and token issuance without manual intervention.
4. **Privacy-Preserving Architecture**
   - Utilizes **Zero Knowledge Proofs (ZKPs)** to enable anonymous, verifiable interactions.
   - Ensures buyers can submit reviews without revealing identity, while proving authenticity.
5. **Incentive Mechanism for Honest Reviews**
   - Issuance of **discount tokens** post-feedback encourages genuine participation.
   - Tokens can be redeemed across multiple platforms, adding tangible value.
6. **Cross-Platform Reputation Portability**
   - Seller reputations are **persistent** and can be shared across multiple marketplaces.
   - Enables a global, unified trust score per seller.
7. **Integration Ready**
   - Designed to be embedded into existing e-commerce ecosystems using **APIs and SDKs**.
   - Avoids the need for building new platforms from scratch.
8. **Scalability and Governance**

- Employs **permissioned blockchain frameworks** like Hyperledger Fabric and Indy for enterprise-grade performance.
- Multi-stakeholder governance ensures collective control, avoiding monopolies.

CHAPTER 3

SYSTEM REQUIREMENTS

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SYSTEM REQUIREMENTS

3.1 Functional Requirements

The Trust chain system is a decentralized reputation management platform for e-commerce environments. It ensures privacy, authenticity, and fairness using blockchain, verifiable credentials, and smart contracts. The key functional requirements are:

➤ **User Identity and Role Management**

- Registration and verification of buyers and sellers using verifiable credentials.
- Role-based access control with distinct capabilities for Buyers, Sellers, and Admins.

➤ **Transaction-Linked Feedback System**

- Feedback tokens issued to buyers upon verified purchase.
- Feedback submissions only permitted if the buyer holds a valid token.

➤ **Discount Token Incentive Engine**

- Automatic issuance of discount tokens to buyers who provide feedback.
- Tokens can be redeemed across supported platforms for rewards or discounts.

➤ **Smart Contract Execution**

- Feedback token validation
- Rating calculation
- Discount token generation
- Immutable storage of review data and transaction logs.

➤ **Reputation Score Management**

- Sellers' reputations are built based on verified feedback history.
- Cross-platform seller identity and reputation tracking enabled by verifiable credentials.

➤ **Administrative Dashboard**

- Monitor system transactions, token flow, and feedback trends.
- View flagged feedback and initiate audits or review fraud detection.

➤ **Security and Access Control**

- Use of zero-knowledge proofs for anonymous yet verifiable interactions.
- Role-based access ensures only authorized users can initiate blockchain transactions.

3.2 Non-Functional Requirements

To ensure robust operation under real-world conditions, the Trust chain system must meet the following non-functional requirements:

➤ **Scalability**

- Modular smart contract architecture for efficient scaling.
- Capable of handling large volumes of transactions across multiple e-commerce platforms.

➤ **Security**

- Use of cryptographic primitives for data integrity.
- Authentication using verifiable credentials and digital signatures.
- Blockchain immutability protects against tampering or unauthorized access.

➤ **Performance**

- Smart contracts optimized for low-gas execution (if on public chain).
- Transactions validated under 2 seconds on permissioned blockchain.

➤ **Privacy**

- Feedback submission through zero-knowledge proofs (ZKP) to ensure user anonymity.
- Selective disclosure of identity attributes.

➤ **Maintainability**

- Smart contracts and token mechanisms deployed modularly for easier upgrades.
- Platform-agnostic APIs for integration with existing e-commerce systems.

➤ **Usability**

- Web-based interface designed for seamless integration with standard online marketplaces.
- Minimal onboarding friction for buyers and sellers.

➤ **Interoperability**

- Compatibility with existing identity standards (e.g., Aadhaar, PAN) via verifiable credentials.
- Seller reputation is portable and valid across multiple platforms.

3.3 System Requirements

3.3.1 Hardware Requirements

Minimum Configuration:

- Processor: Intel Core i3 / AMD Ryzen 3 or above
- RAM: 4 GB
- Storage: 256 GB HDD or SSD
- Display: 15.6" HD Monitor
- Input Devices: Standard Keyboard and Mouse
- Internet: Stable broadband connection (minimum 10 Mbps)

Recommended Configuration:

- Processor: Intel Core i5 / AMD Ryzen 5
- RAM: 8–16 GB
- Storage: 512 GB SSD
- GPU: Integrated graphics sufficient (blockchain nodes are light-weight)
- Display: Full HD Monitor
- Internet: High-speed (20+ Mbps) for consistent node synchronization

Cloud / Deployment Environment:

- VM: Ubuntu 20.04+
- CPU: 2 vCPUs or more
- RAM: 8 GB
- Storage: 100 GB SSD
- Network: Static IP, TLS enabled
- Blockchain node: Hyperledger Fabric/Indy instance or Ethereum private node

3.3.2 Software Requirements

Frontend:

- HTML5, CSS3
- Java Swings / JavaFX (for desktop)
- JavaScript (if web-based UI is enhanced)

Backend:

- JDK 1.8+ with NetBeans 8.2
- Smart Contract Language: Go or Node.js (for Hyperledger), Solidity (for Ethereum)
- RESTful APIs for blockchain interaction.

Blockchain Infrastructure:

- **Hyperledger Fabric**: For permissioned blockchain layer (reputation and tokens)
- **Hyperledger Indy**: For decentralized identity and verifiable credentials
- **Smart Contracts**: Chaincode modules or Solidity contracts

Database:

- MySQL Server 5.0 or above (for off-chain data)
- CouchDB (Hyperledger state DB if applicable)

Tools and DevOps:

- Docker and Docker Compose (for node/container deployment)
- GitHub or GitLab for version control
- Postman for API testing
- JUnit/TestNG for unit testing smart contracts

CHAPTER 4

SYSTEM DESIGN / METHODOLOGY

CHAPTER 4

SYSTEM DESIGN / METHODOLOGY

4.1 Introduction to Software Design

Software design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. For Trustchain, the design phase is critical in ensuring a secure, scalable, and privacy-respecting decentralized application that addresses the shortcomings of centralized reputation systems.

Trustchain employs both high-level architectural design and low-level component interaction to implement a blockchain-powered, verifiable credential-based reputation model for e-commerce.

4.1.1 Design Concepts Applied

The design of Trustchain is grounded in well-established software engineering principles to ensure maintainability, performance, and security. The following design concepts have been applied:

- **Abstraction**: High-level components such as Feedback Tokens, Discount Tokens, and Verifiable Credentials are abstracted into independent modules, simplifying system complexity while enhancing clarity and focus.
- **Refinement**: Core operations—like review submission or reward issuance—are decomposed into smaller tasks such as credential validation, token verification, and smart contract execution, promoting better control and traceability.
- **Modularity**: The system architecture is built from five key modules:
    - Identity Management
    - Token Issuance & Validation
    - Feedback Recording
    - Discount Token Distribution
    - Smart Contract Processing

Each module is logically and functionally independent to support isolated testing, easier updates, and scalability.

- **Software Architecture**: Trustchain follows a decentralized, layered design:
  - User Interface Layer: Java Swing or Web UI for interaction.
  - Middleware Services: Java-based APIs to manage user logic and credential handling.
  - Blockchain Layer: Hyperledger Fabric (for feedback) and Indy (for identity and credentials).
  - Credential Engine: Implements verifiable credential issuance and Zero-Knowledge Proof (ZKP) mechanisms.
- Control Hierarchy: A top-down structure ensures secure and logical execution flow, especially for feedback authentication and reward logic governed by smart contracts.
- Structural Partitioning:
  - Horizontal: Functional segmentation (e.g., authentication, feedback, identity).
  - Vertical: Role-specific access for buyers, sellers, and admins.
- Data Structures: JSON-based credential formats and token schemas are used to represent user identities, transactions, reviews, and scores.
- Software Procedures: Core processes such as token issuance and feedback verification are implemented as discrete, well-defined procedures.
- Information Hiding: Sensitive buyer/seller data is encrypted or shielded from other modules through cryptographic techniques and minimal data exposure, following the principle of least privilege.

4.1.2 Design Considerations

The design of Trustchain reflects practical and technical factors essential to ensuring system quality and usability:

- Compatibility: Designed for seamless API-based integration with existing e-commerce platforms, allowing real-world adoption without the need for architectural overhauls.
- Extensibility: The modular architecture allows easy introduction of new features, such as additional token types or trust metrics, without disrupting existing workflows.
- Fault-Tolerance: Operates on distributed permissioned blockchains, ensuring resilience against

individual node failures and preventing system downtime.

- Maintainability: Well-separated modules and services allow localized updates and bug fixes, minimizing impact on other components.
- Modularity & Reusability: Modules such as the credential engine or feedback token validator are built for reuse across different projects or platforms that require verifiable reputation systems.
- Reliability: Every review is cryptographically validated and stored immutably, ensuring consistent, tamper-proof data integrity across the system.
- Robustness: Includes safeguards to handle expired tokens, duplicate submissions, invalid identities, or misuse attempts without system breakdown.
- Security: Uses blockchain consensus, smart contracts, and verifiable credentials to enforce trust, prevent Sybil attacks, and guarantee data authenticity.
- Usability: A simplified UI guides buyers and sellers through feedback, review, and reward processes, ensuring ease of use while maintaining full system functionality.

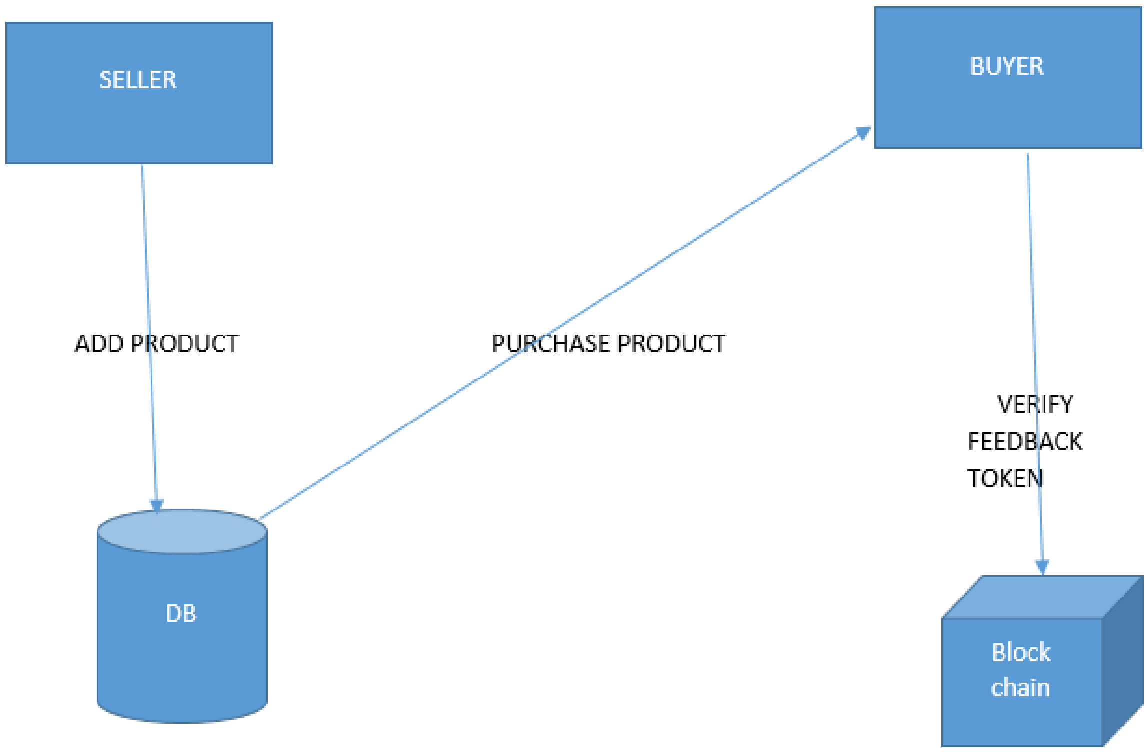

Figure 4.1 – System Architecture

4.2 Data Flow Design

A **data-flow diagram** (**DFD**) is a graphical representation of the "flow" of data through an information system. DFDs can also be used for the visualization of data processing (structured design).

On a DFD, data items flow ***from*** an external data source or an internal data store ***to*** an internal data store or an external data sink, ***via*** an internal *process*.

A DFD provides no information about the timing or ordering of processes, or about whether processes will operate in sequence or in parallel. It is therefore quite different from a flowchart, which shows the ***flow of control*** through an algorithm, allowing a reader to determine what operations will be performed, in what order, and under what circumstances, but ***not*** what kinds of data will be input to and output from the system, ***nor*** where the data will come from and go to, ***nor*** where the data will be stored (all of which are shown on a DFD).

4.2.1. DFD Level 0 – Context Diagram

This level illustrates the overall interaction of the TrustChain system with external entities:

- **Buyers** submit feedback through verifiable credentials.
- **Sellers** are assigned unique digital identities and receive reputation updates.
- **E-commerce platforms** issue credentials and host feedback portals.

- **Permissioned Blockchain Nodes** manage token issuance and data recording.

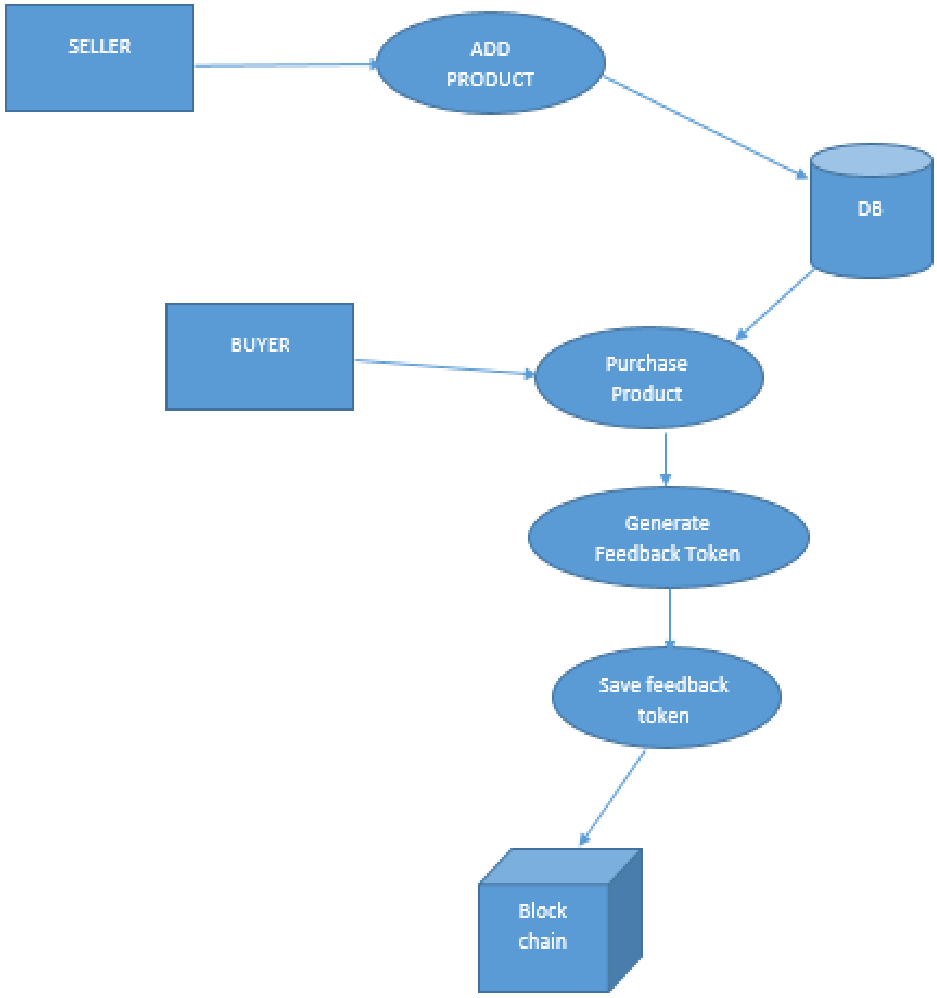

Figure 4.2 – Level Zero DFD

4.2.2. DFD Level 1 – Main Functional Processes

This diagram decomposes the system into the following core functions:

- **Credential Issuance**: After a transaction, platforms issue feedback tokens to verified buyers.
- **Feedback Submission**: Buyers submit feedback along with their feedback tokens.
- **Reputation Update**: The system calculates and updates the seller's reputation score using smart contracts.
- **Reward Distribution**: Honest feedback submission is incentivized with discount tokens, traceable and redeemable.

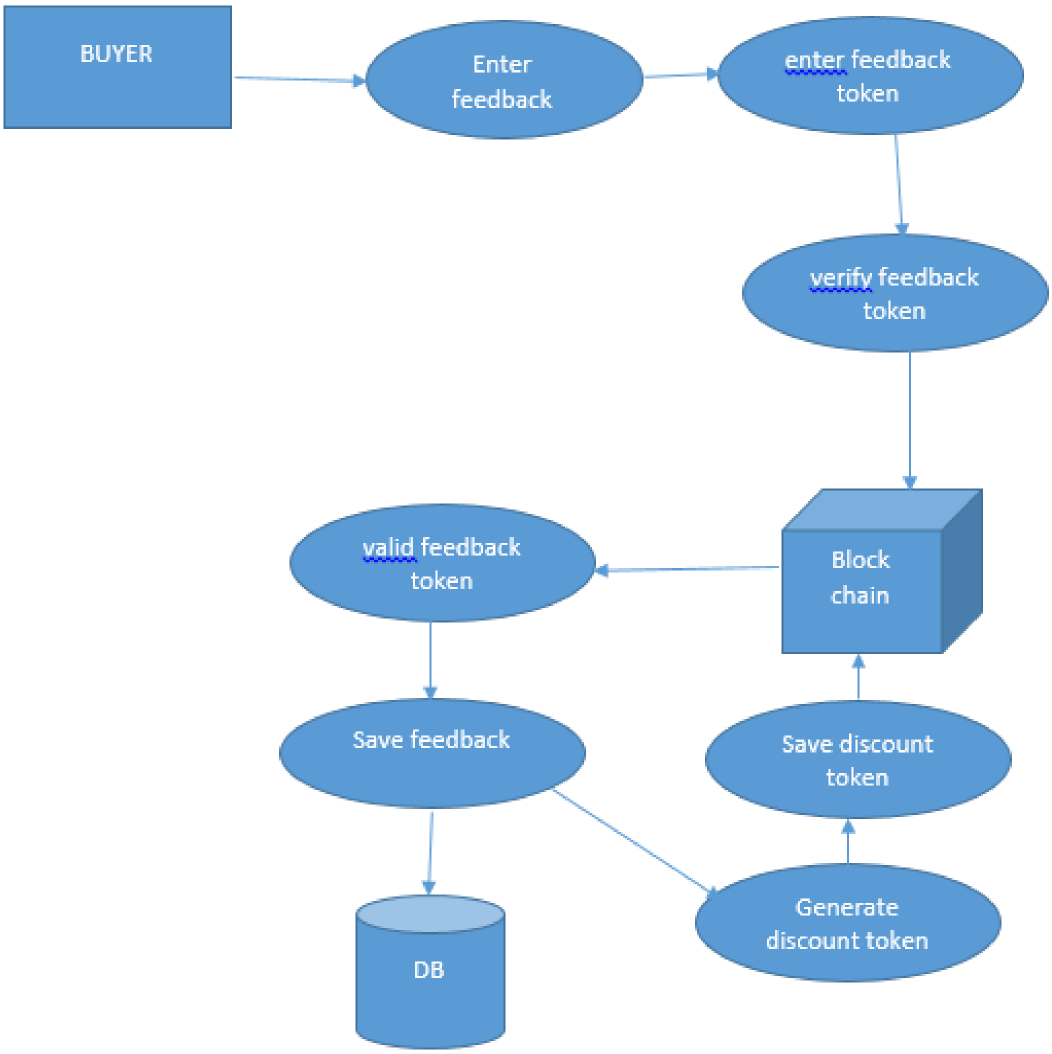

Figure 4.3 – Level One DFD

4.2.3. DFD Level 2 – Feedback Processing Subsystems

This level provides a detailed look into feedback processing:

- **Credential Verification Module**: Confirms the validity and issuer of the verifiable credential using a Verifiable Data Registry.
- **Sybil Attack Filter**: Detects multiple identities using digital fingerprinting and blocks suspicious clusters.
- **Smart Contract Validator**: Runs logic to verify that only eligible feedback (matching completed transactions) is recorded.
- **Reputation Ledger Update**: Writes the feedback hash and rating to the blockchain, ensuring immutability.

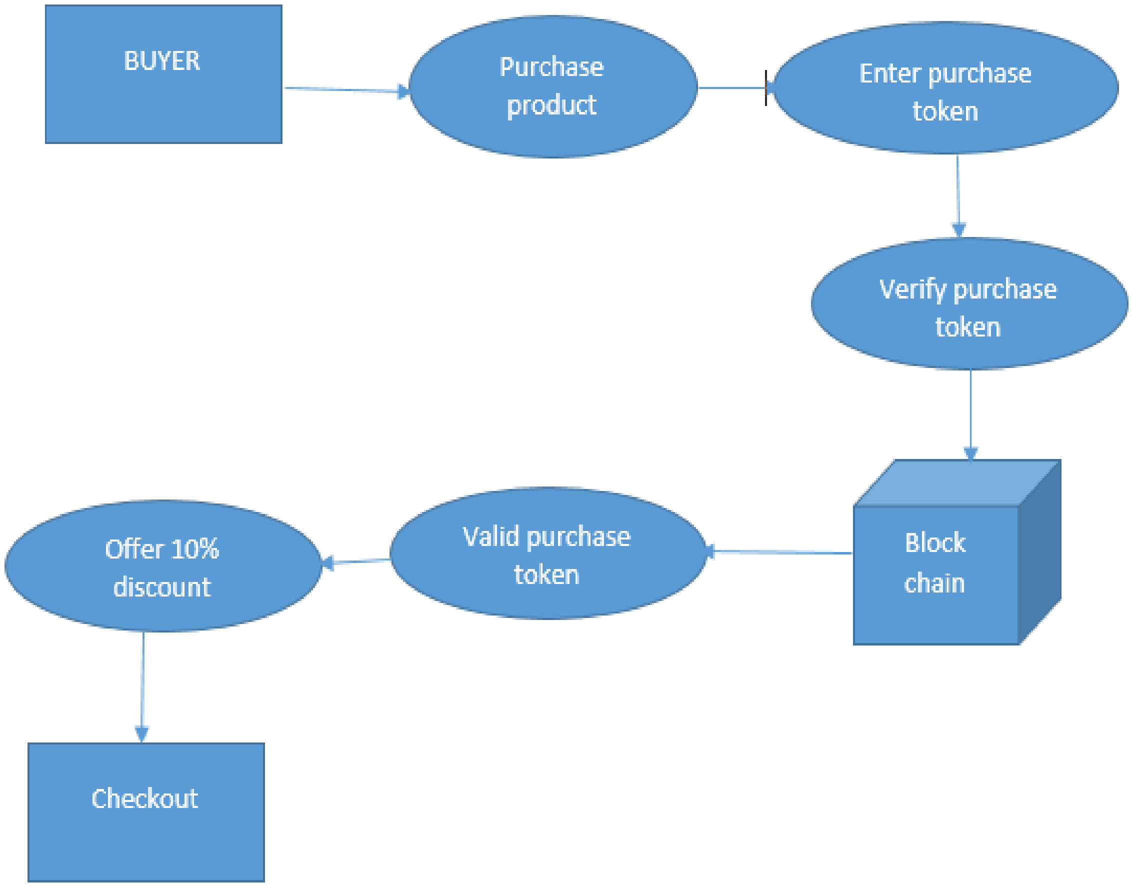

Figure 4.4 – Level Two DFD

4.3 Methodology

The methodology for implementing the TrustChain system is based on a step-by-step modular approach combining blockchain, smart contracts, and cryptographic identity management.

Step 1: Requirement Gathering and Analysis

- Studied existing centralized reputation systems and documented vulnerabilities (e.g., Sybil attacks, fake reviews).

- Identified critical system requirements including privacy preservation, decentralized trust, and cross-platform operability.

Step 2: Identity Framework Design

- Integrated **Verifiable Credentials (VCs)** following W3C standards.

- Buyers receive **Feedback Tokens** post-purchase, tied cryptographically to a transaction.
- Sellers are registered with **Digital Seller Credentials** (e.g., Aadhaar-based or domain-validated identities).

Step 3: Blockchain Architecture

- Employed **Two Permissioned Blockchains**:
  - One for **feedback recording** and credential verification.
  - Another for **reward token issuance and tracking**.
- Each block stores encrypted feedback entries and identity proofs.

Step 4: Smart Contract Development

- Developed smart contracts to:
  - Validate feedback submission using zero-knowledge proofs (ZKPs).
  - Compute and update seller scores based on predefined algorithms.
- Automatically issue **Reward Tokens** to buyers.

Step 5: Feedback Validation Logic

- Implemented the following logic:
  - Each feedback token must be single-use and traceable to a real transaction.
  - Only credentialed buyers can submit ratings.
  - Smart contract rejects reused, expired, or fake tokens.

Step 6: Frontend & API Integration

- Created a decentralized interface using React.js or similar framework.
- API gateways built using Node.js handle off-chain user requests and synchronize with on-chain state.

Step 7: Testing and Simulation

- Ran unit and integration tests using tools like Truffle and Ganache.
- Simulated attacks (e.g., Sybil and ballot-stuffing) to test resilience and response accuracy.

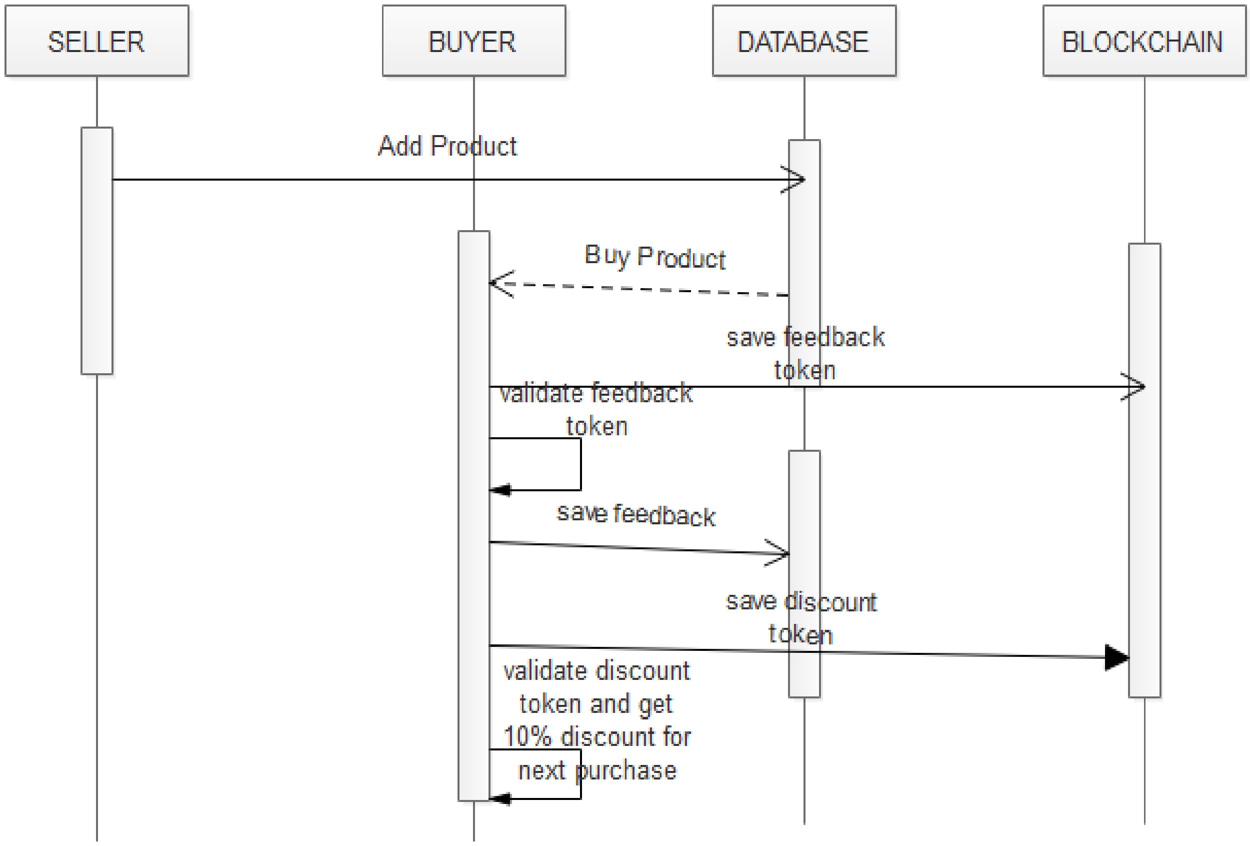

Figure 4.5: Sequence Diagram

CHAPTER 5

IMPLEMENTATION

CHAPTER 5

IMPLEMENTATION

5.1 OVERVIEW

This chapter thoroughly explains the end-to-end implementation of **TrustChain**, a blockchain-powered e-commerce reputation system designed to enhance transparency, trustworthiness, and reliability within online marketplaces. The primary goal of this system is to enable secure, tamper-proof collection and verification of buyer feedback and seller reputation, addressing key challenges such as fraudulent reviews, feedback manipulation, and lack of transparency in traditional e-commerce platforms.

The TrustChain implementation leverages blockchain technology for immutable and verifiable storage of feedback and discount tokens, ensuring that user-generated reputations are authentic and resistant to tampering. The system integrates a modular approach comprising seller and buyer registration and authentication, product listing and purchase, feedback management, and a decentralized blockchain server that securely stores cryptographic tokens related to transactions and feedback.

The backend services are developed using Java with socket programming to facilitate real-time interaction between clients and the blockchain server. The blockchain server, implemented using Java Swing and low-level networking protocols, constructs blocks containing feedback tokens and discount tokens, thereby ensuring a decentralized ledger that can be independently audited for authenticity. This prevents malicious users from falsifying feedback or rewards and supports a trustworthy marketplace ecosystem.

To begin the implementation, sellers and buyers first register and authenticate themselves on the platform. Sellers can add product listings with details including descriptions, prices, and images, while buyers browse these listings and place orders. Upon successful purchase, a unique feedback token is generated and sent to the buyer's registered email, while an identical token is stored on the blockchain ledger for later verification.

Buyers provide feedback only after validating the authenticity of their feedback token, which prevents fake or unauthorized reviews. Upon successful feedback submission, buyers receive discount tokens, again stored both in the system database and on the blockchain, enabling secure and verifiable reward mechanisms. This dual-token system ensures transparency and builds user trust in the fairness of the reputation and rewards system.

The design of TrustChain focuses on scalability and modularity, enabling future enhancements such as integration with third-party delivery systems, real-time order tracking, and advanced analytics on reputation trends. The entire implementation ensures that the system is resilient, secure, and user-friendly, making blockchain technology accessible and practical for everyday e-commerce users.

In summary, this implementation chapter details how blockchain technology, combined with traditional client-server architecture, provides a robust framework to prevent feedback fraud, enhance user trust, and foster a reliable reputation system in e-commerce environments.

5.2 ENVIRONMENT SETUP

The implementation of the **TrustChain: Blockchain Powered E-Commerce Reputation System** required a comprehensive development environment capable of supporting blockchain functionalities, secure authentication mechanisms, real-time token issuance, and a user-friendly interface. The chosen tools and technologies were selected based on their performance, scalability, interoperability, and suitability for both web and desktop components.

Programming Languages and Technologies

- **Java**: Used extensively for building backend services and the blockchain server. Java’s platform independence and robust networking libraries made it ideal for socket programming, credential validation, and blockchain logic. It provided the core infrastructure for client-server interactions and smart contract execution.
- **Java Swing**: Used for the blockchain server’s graphical interface. It provides a visual representation of the blockchain, enabling administrators to monitor token-related transactions, track block additions, and ensure the integrity of feedback entries in real time.
- **HTML/CSS/JavaScript**: Utilized to design responsive and interactive web interfaces for both buyers and sellers. These technologies power the login system, product dashboards, and feedback submission modules, ensuring cross-browser compatibility and a seamless user experience.
- **MySQL**: A relational database used to store structured data such as user credentials, product details, purchase orders, and verified feedback entries. The database acts as a bridge between the blockchain server and the web interface.
- **JavaMail API (SMTP)**: Used for securely sending feedback and discount tokens to users via email. This ensures that only legitimate buyers can access feedback functions, acting as a two-

step verification mechanism.

Development Tools

- **Eclipse / IntelliJ IDEA**: These integrated development environments (IDEs) were employed for writing, debugging, and managing Java code efficiently. Their comprehensive support for code refactoring, version control, and plugin integration streamlined the backend development process.
- **XAMPP**: A cross-platform web server package used to host the MySQL server and test the integration of backend scripts with the database. It enabled efficient simulation of server interactions on localhost.
- **Postman**: A REST API client used extensively during development and debugging to test communication between frontend components and backend services.

Hardware Requirements

- A development machine with a minimum of **8GB RAM**, **multi-core processor**, and a stable internet connection was required.
- **Localhost configuration** was used for socket-based communication between the blockchain server (backend) and the client modules (frontend), emulating real-time interaction in a controlled environment.

5.3 Modules Implementation

The system comprises multiple functional modules, each implementing a core process of the e-commerce platform using blockchain-enhanced trust mechanics.

5.3.1 Seller Registration and Login

Sellers begin by registering with their personal and contact details, including a secure password. These credentials are validated, hashed, and stored securely in the MySQL database. Upon successful login using valid credentials, sellers gain access to a personalized dashboard that allows them to manage product listings, monitor sales, and track buyer feedback. The login system enforces secure authentication and protects user sessions through token-based access.

5.3.2 Seller Product Management

Sellers can add new products by submitting key details such as product name, description, price, quantity, and uploading cover images. Each product entry is stored in the database and made publicly viewable for buyers. Sellers can also review and manage their active listings, view purchase history, and update product availability in real time.

5.3.3 Buyer Registration and Login

Buyers register similarly with their details, creating a secure account in the system. Login is authenticated through hashed credentials, allowing buyers to access features such as product browsing, purchase history, and feedback submission. This process ensures a secure environment for buyers to interact with the platform and maintain accountability.

5.3.4 Product Browsing and Purchase

All listed products are presented in a centralized catalog with detailed information including seller name, product price, and available stock. Upon selecting and purchasing a product, the system automatically generates a **feedback token** tied to the buyer’s email. This token is also recorded immutably on the blockchain, ensuring that feedback can only be submitted by verified purchasers and cannot be fabricated or reused.

5.4 Feedback and Token Management

A crucial innovation in TrustChain is the issuance and management of tokens for feedback submission and reward incentives.

5.4.1 Feedback Submission

After receiving a product, the buyer is prompted to submit feedback. This process begins by validating the **feedback token** sent to the buyer’s email. The token is cross-verified against blockchain records to ensure authenticity, prevent duplication, and validate transaction legitimacy. Upon verification, the feedback is saved in the database and linked to the relevant product and seller, contributing to their reputation score.

5.4.2 Discount Token Generation

Once feedback is successfully submitted and validated, the system issues a **discount token** to the buyer as an incentive for honest participation. This token is both emailed to the buyer and recorded on the blockchain, making it tamper-proof and reusable only under legitimate conditions. Buyers can apply this token for future purchases, thereby fostering continued engagement and loyalty within the platform.

5.5 Blockchain Server Implementation

The blockchain server is central to TrustChain's decentralized trust model. Developed using **Java Swing**, the server provides both functional and visual components.

- The GUI allows administrators to monitor real-time transactions, visualize the blockchain structure, and verify the integrity of stored blocks.
- Using **socket programming**, the server listens to requests from clients (i.e., the frontend modules) such as feedback validation or token issuance.
- Each feedback and discount token is recorded as a new block in the chain. These blocks are cryptographically linked, ensuring immutability—once a block is added, it cannot be altered without invalidating the entire chain.
- The server ensures decentralized verification by acting as an authority-free ledger that stakeholders (e.g., platforms or sellers) can independently audit for fairness and transparency.

5.6 System Security and Data Integrity

Security and integrity are foundational to TrustChain's design, ensuring that feedback cannot be forged and user data remains protected at all times.

- **Credential Security**: All user passwords are hashed before being stored in the database, preventing credential leaks in case of data breaches.
- **Token Security**: Feedback and discount tokens are generated using secure, tamper-resistant algorithms. They are further safeguarded by storing them on the blockchain, where they cannot be modified or reused fraudulently.

- **Blockchain Immutability**: Each transaction is permanently written to the ledger. This ensures that neither users nor administrators can alter feedback or tokens post-submission.
- **Two-Factor Verification**: Feedback tokens are delivered via email and cross-verified with blockchain records before acceptance, acting as a dual layer of validation.
- **Secure Communication**: Socket communication between the blockchain server and client modules is encrypted to prevent unauthorized access or interception by malicious actors.
- **Auditability and Backups**: All blockchain activity is logged and periodically backed up, allowing for transparency, traceability, and disaster recovery in case of system failure.

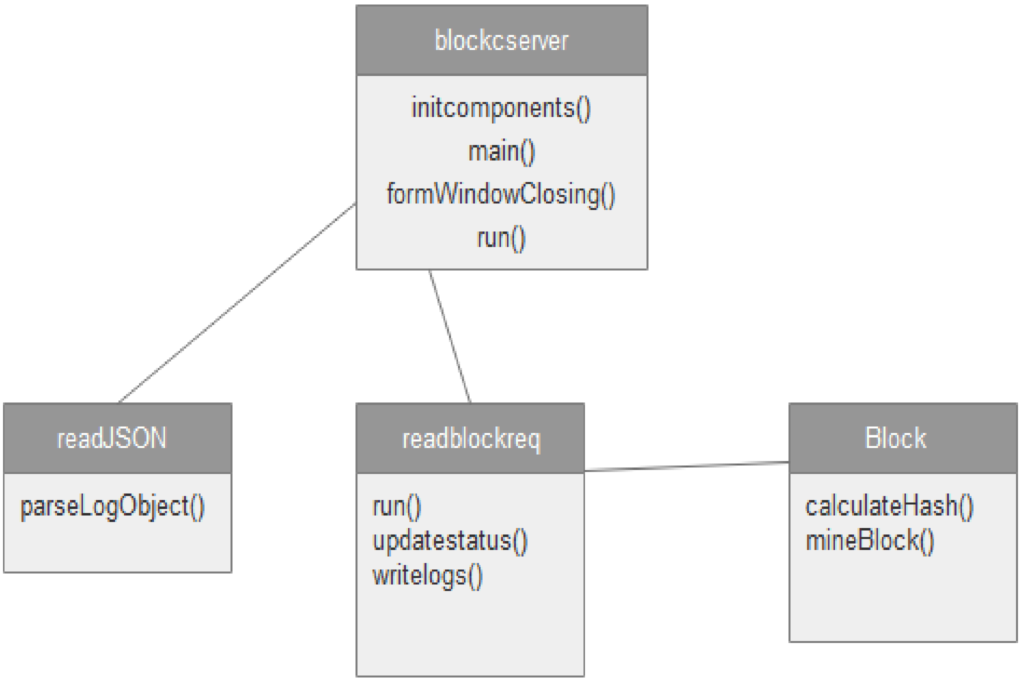

Figure 5.1 – Structure of React Application

CHAPTER 6
TESTING

CHAPTER 6

TESTING

6.1 Testing

Software testing is a critical and indispensable phase in the Software Development Life Cycle (SDLC). It serves as the final checkpoint before deployment, ensuring that the software product meets its defined functional and non-functional requirements. Testing is carried out to identify defects, confirm that the software behaves as expected under various scenarios, and verify that it performs reliably, securely, and efficiently in real-world conditions.

In the context of the Trustchain system, testing plays a pivotal role due to the involvement of decentralized technologies, cryptographic verifications, and blockchain interactions. The system incorporates complex modules such as verifiable credentials, smart contracts, token-based rewards, and immutable feedback storage, all of which must function correctly in coordination.

This chapter describes the testing strategy adopted to validate the Trustchain platform. It outlines the various types of testing performed—ranging from unit testing to user acceptance testing—and demonstrates how each level of testing contributed to ensuring the accuracy, integrity, and robustness of the system. The goal was not only to catch programming or logic errors but also to validate security protocols, integration between components, and the overall user experience.

6.1.1 Aim of Testing

The primary aim of testing in the Trustchain: Blockchain Powered E-Commerce Reputation System is to ensure that the system functions as intended and meets user, functional, and technical requirements. Given its integration with blockchain and cryptographic verification, the goal is to detect errors in token handling, credential validation, and system behavior before deployment.

Testing also ensures that:

- Feedback and discount tokens are issued and verified correctly.
- Interactions with the blockchain are secure and consistent.
- All modules work reliably together under expected conditions.
- The system handles invalid input and misuse scenarios gracefully.

6.1.2 Testing Principles

Testing for Trustchain was guided by key industry principles:

- Early Testing: Initiated during development to catch issues early.
- Focused Testing: Emphasis on critical components like feedback validation and smart contracts.
- Effective Coverage: Prioritized meaningful test cases over exhaustive combinations.
- Defect Detection: Aimed to identify and fix flaws rather than prove system perfection.
- Repeatable and Documented: Tests were structured, reusable, and results were logged for traceability.
- User-Oriented: Final testing phases focused on real user scenarios and expectations.

6.1.3 Test Plan

The Trustchain test plan outlines how the system was tested across different layers and user flows.

- Scope: Testing of login, product purchase, feedback submission, token issuance, and reward redemption.
- Test Environment: Java backend, blockchain nodes (Hyperledger Fabric & Indy), MySQL database, and frontend UI.
- **Key Modules Tested:**
  - User authentication
  - Token validation and storage
  - Feedback recording and reputation update
  - Discount token generation and application

- **Team Involvement:**
  - Developers handled unit tests
  - Testers managed functional and integration testing
  - End-users contributed to acceptance testing

- **Criteria:**
  - Entry: Completion of module integration
  - Exit: All major features tested and approved, with no critical defects

- **Deliverables**: Test reports, bug logs, and UAT approval

6.2 Type of Testing

6.2.1. Unit Testing

Unit testing verifies that individual components (e.g., feedback validation, token generation) operate as intended. These tests are performed at the code level for core logic and smart contract modules, ensuring correct inputs produce the expected outputs. Unit tests were written for:

- Token validation logic
- Credential issuance and verification
- Smart contract functions like validateToken() and issueRewardToken()

6.2.2. Functional Testing

Functional tests validate that the features of the Trustchain system meet business and user requirements. These tests ensure:

- Valid inputs are accepted, and invalid inputs are handled gracefully
- All core functions like product listing, purchase, feedback, and rewards behave correctly
- Feedback and discount tokens are processed securely and accurately

Functional test cases included:

- Login authentication for buyers/sellers
- Adding products to the system
- Buyer product purchase and feedback flow
- Discount token redemption

6.2.3. System Testing

System testing verifies the behavior of the entire integrated Trustchain platform. It ensures all modules (blockchain backend, database, frontend) work together as a unified application. It checks:

- End-to-end flow from buyer purchase to feedback submission and token reward
- Interoperability between components like smart contracts and verifiable credentials
- System behavior under typical usage scenarios

6.2.4. Integration Testing

Integration testing checks the data flow and interaction between modules, especially:

- Communication between API server and blockchain network

- Token validation and credential issuance on-chain
- Email notifications and buyer verification

Integration checks included:

- IP communication with blockchain node
- Product persistence in database
- Email confirmation for buyers
- Token state validation post-submission

6.2.5. Performance Testing

Performance tests evaluate system response time and resource handling under simulated load conditions. Tests were designed to verify:

- Response time for review submission and blockchain confirmation
- System performance under simultaneous user access
- Token issuance latency

6.2.6. Acceptance Testing

User Acceptance Testing (UAT) was conducted with selected end-users to validate that the system meets real-world expectations and business requirements. Users tested login flows, product purchase, feedback submission, and reward redemption. All major use cases were successfully validated.

Table 6.1 – Unit testing Test cases

Test Case	Module	Given Input	Expected Output	Actual Output	Result
1	Seller/Buyer Login	Username & Password	User has to be authenticated and allowed to login	User successfully authenticated and logged in	Pass
2	Add Product	Product Details	Product details to be saved in DB	Product details successfully saved in DB	Pass

3	View Orders	Username	All orders of user should be displayed	All orders successfully displayed	Pass

Table 6.2 – Integration testing Test cases

Test Case	Module	Given Input	Expected Output	Actual Output	Result
1	Buy Product (Buyer)	Product Details	Product should be purchased and feedback token saved in blockchain; email sent	Feedback token successfully saved in blockchain and mail sent to buyer	Pass
2	Enter Feedback	Feedback token and feedback	Token should be verified in blockchain; feedback saved; discount token generated	Token verified in blockchain, feedback saved, discount token generated & saved	Pass

Table 6.3 – System testing Test cases

Test Case	Module	Given Input	Expected Output	Actual Output	Result
1	Buy Product with Discount	Product details and discount token	Discount token should be verified and 10% discount given on the product	Discount token verified and 10% discount given on the product	Pass

CHAPTER 7
RESULTS

CHAPTER 7

RESULTS

This section showcases the main home page of the Trustchain System, where it has all the seller and buyer details page.

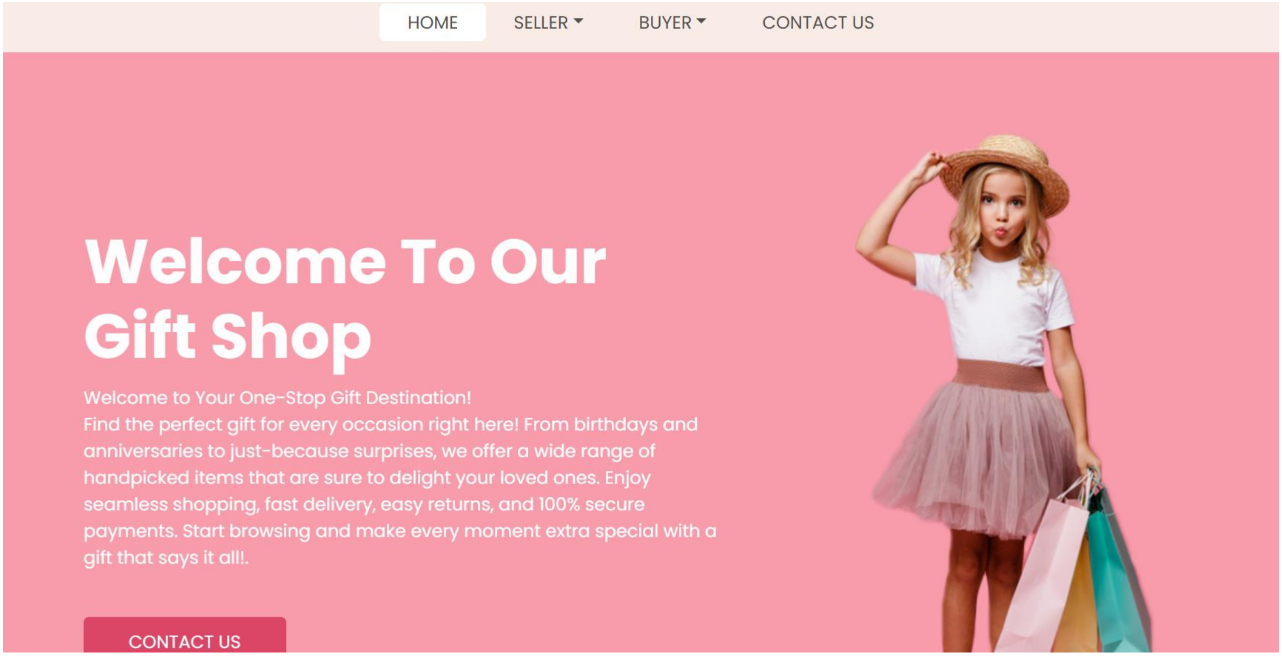

Figure 7.1: Home Page

Figure 7.1 represents the home page of the Trustchain-powered e-commerce platform, designed to offer a seamless and role-based entry point for users. At the top of the interface, users are provided with clear options to navigate as either a Seller or a Buyer, with an additional "Contact Us" section available for support and general inquiries. This role selection is crucial as it dynamically adjusts the user experience and backend services according to the selected profile. The login interface that follows allows users to securely enter their registered phone number and password, with an optional "Remember Me" feature for convenience. Integrated with the authentication microservice, the system validates credentials securely before granting access through token-based login, ensuring both privacy and system integrity.

Below the navigation and login elements, the page transitions into a modern, responsive product showcase—visually similar to popular e-commerce platforms. This section displays a catalog of available items including product names, images, prices, available quantities, and seller information. Each product is accompanied by functional options such as "Buy Now" or "View Details," encouraging user engagement and facilitating direct interaction between buyers and sellers. The product listings are dynamically rendered from the backend database in real time, ensuring that users always view the most up-to-date inventory. This comprehensive home page serves not only as a login portal but also as a transparent and user-friendly marketplace that emphasizes accessibility, trust, and usability—aligned with the goals of a decentralized, reputation-driven commerce environment.

Figure 7.2: The Blockchain Server Page

The Blockchain Server Page operates as the core backend component of the TrustChain system, responsible for managing all token-related transactions between buyers and sellers. Running continuously in the background, this server handles the issuance, verification, and recording of both **feedback tokens** and **discount tokens**. Built using Java Swing for GUI visualization and socket programming for real-time communication, it maintains an **immutable ledger** where each token event—whether generated, validated, or redeemed—is stored as a cryptographically linked block. This ensures data integrity, prevents tampering, and allows for independent verification of reputation-related actions across the platform. The server plays a pivotal role in enforcing trust, transparency, and security throughout the system.

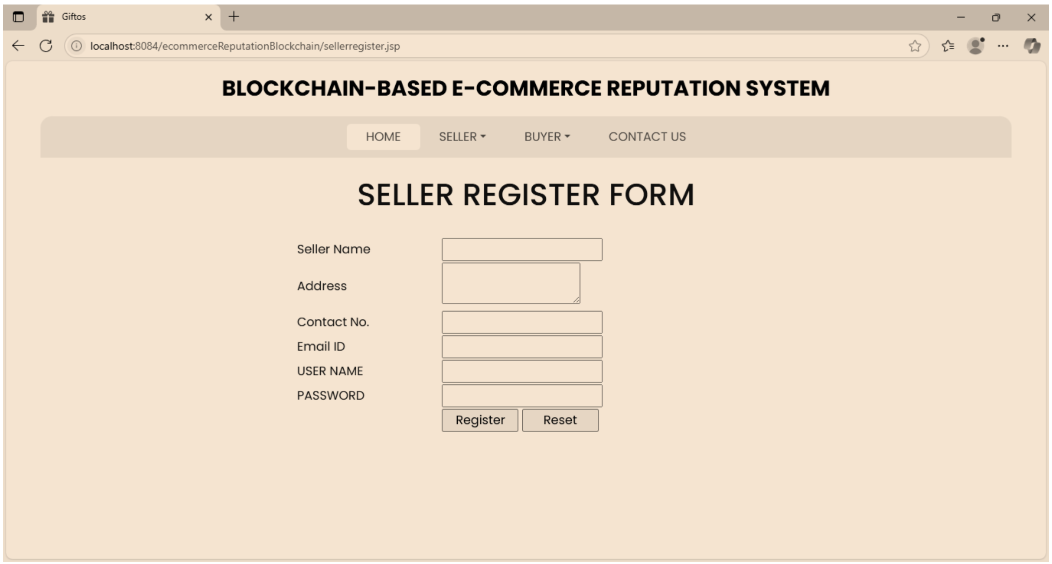

Figure 7.3: The Seller Registration Form

The Seller Registration Form is designed to securely capture essential information required to onboard new sellers onto the TrustChain platform. The form includes fields for **Seller Name**, **Address**, **Contact Number**, **Email ID**, **Username**, and **Password**. All inputs are validated to ensure accuracy and completeness. The password is securely hashed before being stored in the database, protecting user credentials from unauthorized access. Upon successful registration, the seller gains access to a personalized dashboard to manage products, view orders, and monitor feedback, ensuring a trusted and authenticated seller identity within the system.

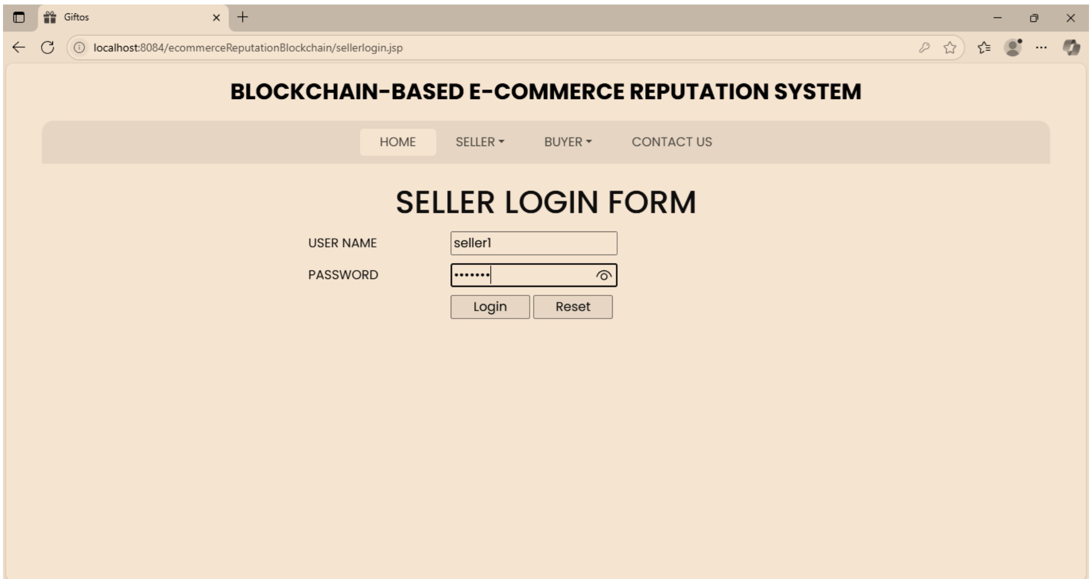

Figure 7.4: The Seller Login Form

The Seller Login Form provides a secure entry point for registered sellers to access their accounts. It includes fields for **Username** and **Password**, along with **Login** and **Reset** buttons. Upon entering valid credentials, the seller is authenticated and granted access to their dashboard, where they can **add products**, manage listings, and track orders. If incorrect information is entered, the form prompts appropriate error messages, ensuring secure and user-friendly access control.

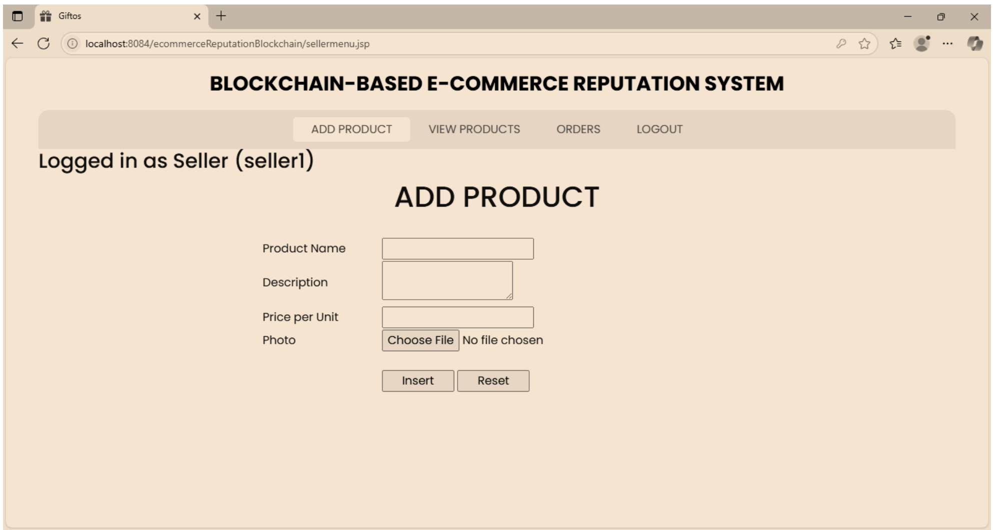

Figure 7.5: Add Product

The Add Product Page allows authenticated sellers to list new items for sale on the platform. The form includes fields for **Product Name**, **Blockchain Type** (used to tag the product within the TrustChain system), **Price per Unit**, and an option to **upload a product photo**. Once submitted, the product details are stored in the database and become visible to buyers on the storefront. This page ensures a simple and efficient product listing process while maintaining secure linkage to the seller's verified blockchain identity.

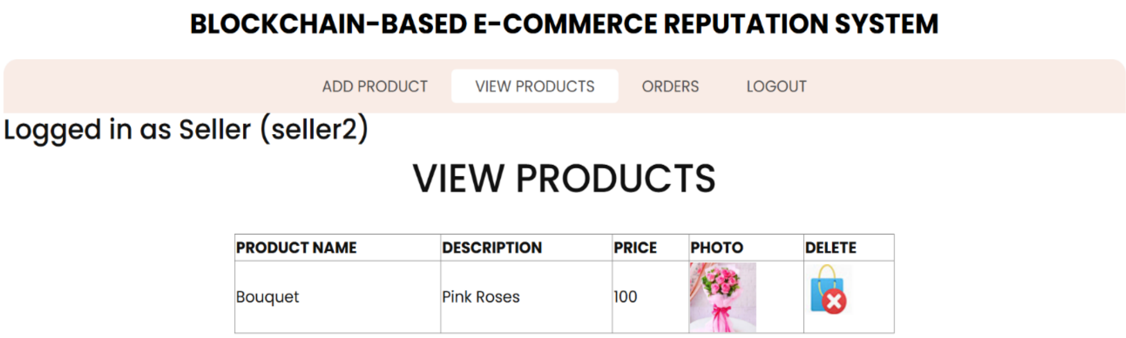

Figure 7.6: Add Product

The View Products Page provides sellers with a comprehensive overview of all products they have listed on the platform. Displayed in a tabular format, the page includes columns for **Product**

Name, **Description**, **Price**, **Product Photo**, and a **Delete Option**. Sellers can easily review their inventory, update product details, or remove items as needed. This feature offers a centralized and user-friendly interface for managing product listings efficiently within their own seller account.

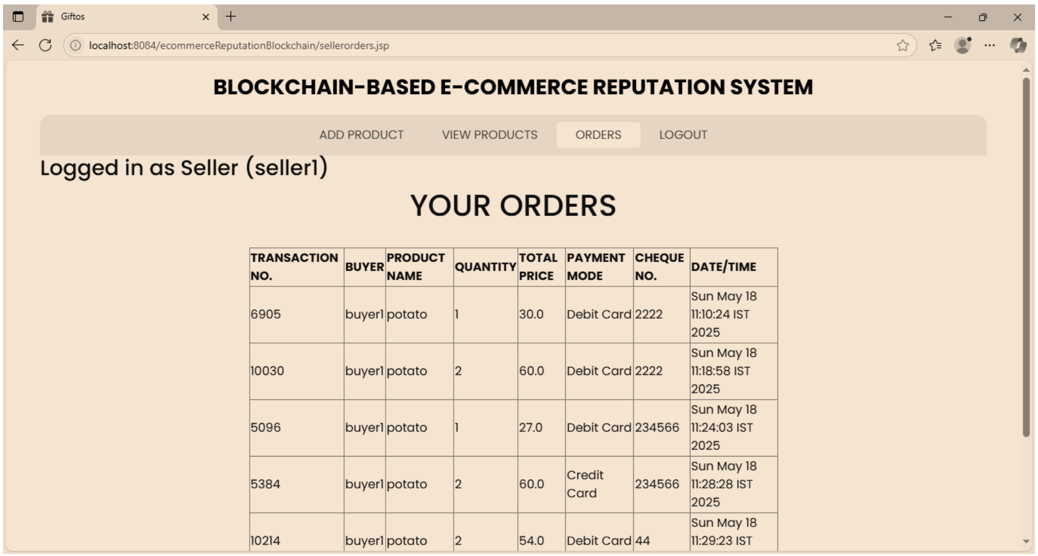

Figure 7.7: The Orders Page (Seller)

The Orders Page allows sellers to view and manage all incoming orders placed by buyers for their listed products. Displayed in a detailed table, the page includes key order information such as **Transaction Number, Buyer Name, Product Name, Quantity Ordered, Total Price, Payment Mode, Cheque Number** (if applicable), and Date/Time of the transaction. This interface helps sellers track sales, verify payment details, and maintain an organized record of transactions, enabling smooth and transparent order fulfillment.

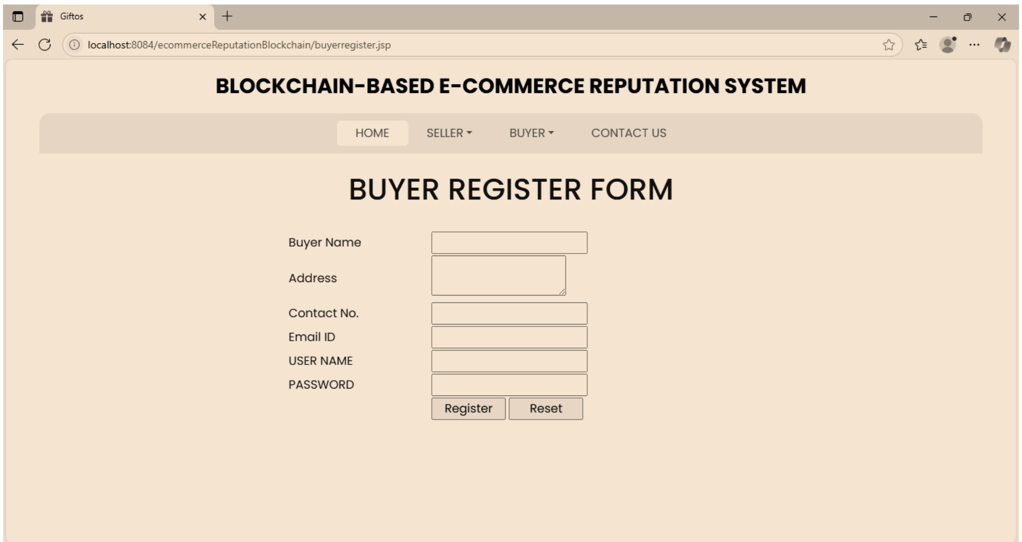

Figure 7.8: The Buyer Registration Form

The Buyer Registration Form is designed to securely collect necessary details from new users who wish to make purchases on the platform. It includes input fields for **Buyer Name**, **Address**, **Contact Number**, **Email ID**, **Username**, and **Password**. All fields are validated to ensure correct and complete information. The password is securely hashed before storage to maintain user privacy. Upon successful registration, the buyer gains access to their personalized dashboard, enabling them to browse products, place orders, and submit feedback with blockchain-verified trust.

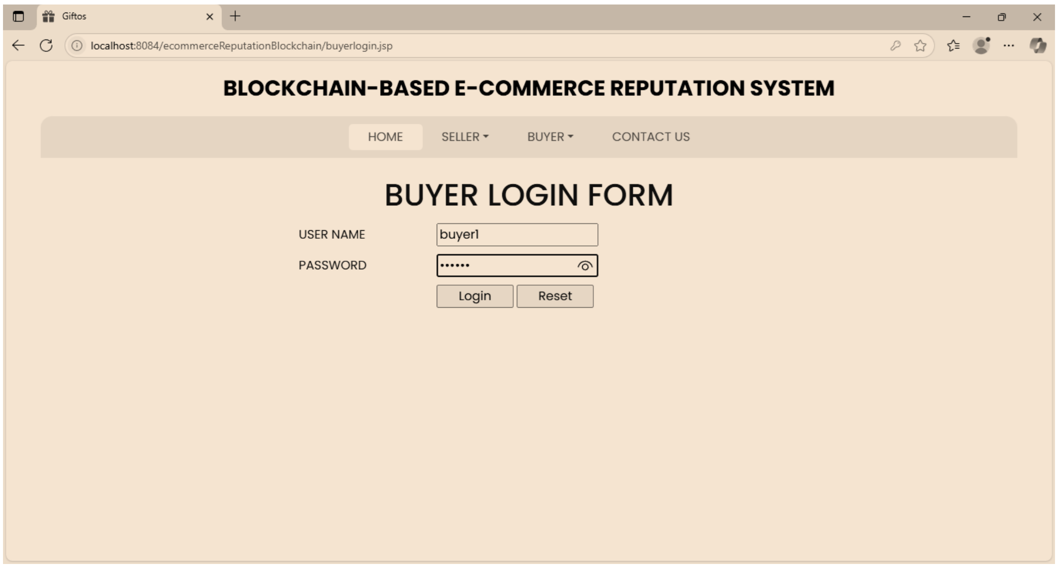

Figure 7.9: The Buyer Login Form

The Buyer Login Form provides a secure gateway for registered buyers to access their account. It includes fields for **Username** and **Password**, along with **Login** and **Reset** buttons. Upon successful authentication, the buyer is directed to their dashboard where they can **browse products**, **place orders**, and **submit feedback**. The form ensures only authorized users can access buyer-specific functionalities, maintaining the integrity and privacy of the platform.

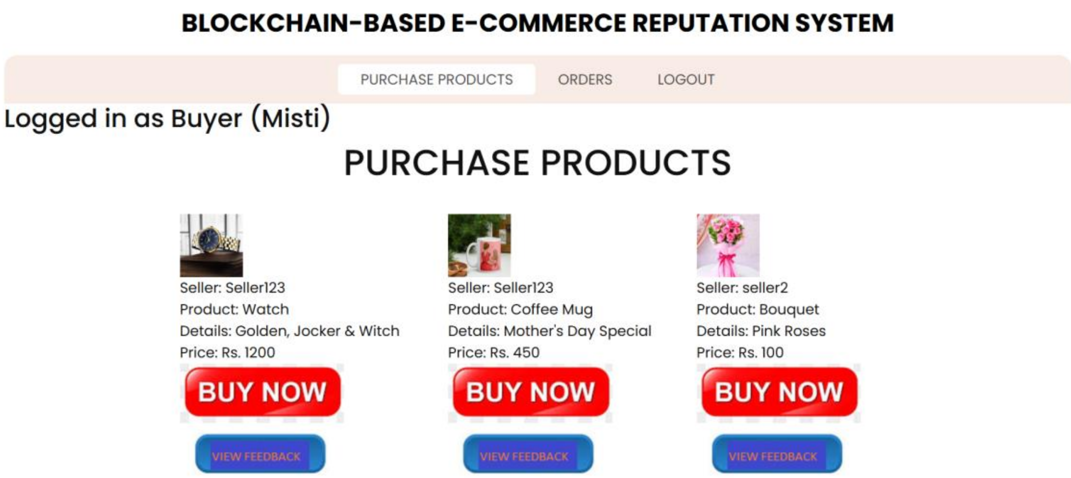

Figure 7.10: The Purchase Products Page

The Purchase Product Page allows buyers to view complete details of a selected product, including

Product Name, Description, Price, Seller Information, and **Product Image**. From this page, the buyer can confirm the quantity and proceed to **purchase the produc**t. Once the purchase is completed, a **feedback token** is generated and sent to the buyer's email, initiating the feedback and reward cycle. This page ensures a smooth, transparent, and user-friendly buying experience within the Trust Chain platform.

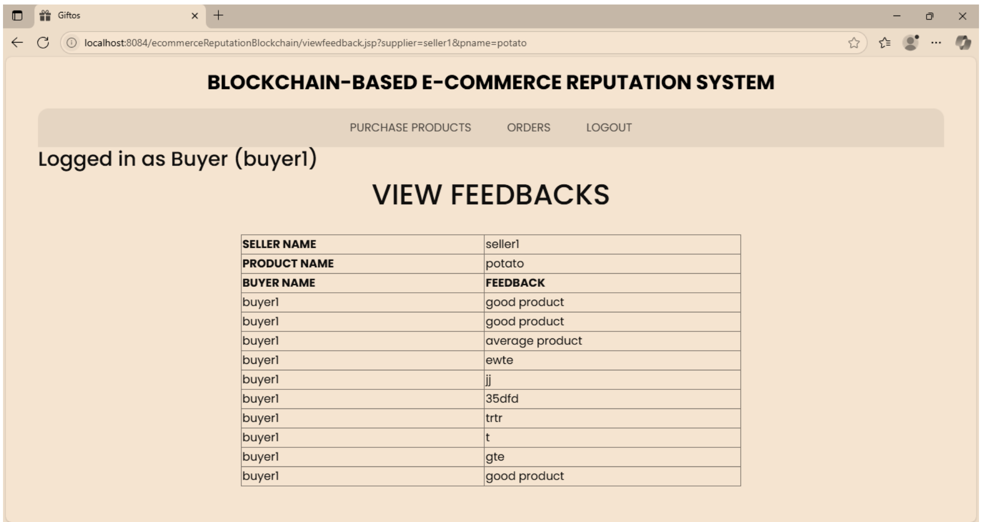

Figure 7.11: The Purchase Products Page

The View Feedback Page enables buyers to review their past purchases and submit feedback for products they have bought. It displays details such as **Product Name**, **Seller Name**, **Purchase Date**, and **Feedback Status**. Buyers can enter their feedback using the **valid feedback token** received via email. Once submitted, the feedback is securely recorded and linked to the corresponding product and seller, contributing to their reputation on the blockchain. This page promotes transparency and encourages honest user participation.

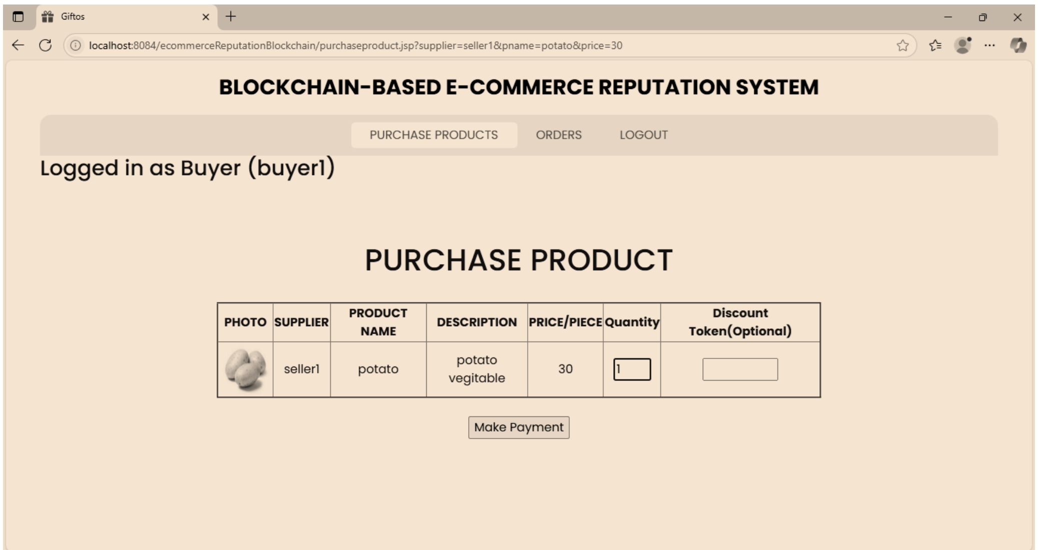

Figure 7.12: The Purchase Products Page (to make payment)

The Purchase Product Page allows buyers to select and buy products by viewing all essential details in a structured table format. The table includes the **Product Photo**, **Supplier Name**, **Product Name**, **Description**, **Price**, **Quantity**, and an optional field for entering a **Discount Token**. Buyers can specify the quantity they wish to purchase, apply a valid discount token if available, and proceed to make the payment. This page ensures a transparent, secure, and user-friendly purchasing experience, streamlining the entire buying and checkout process.

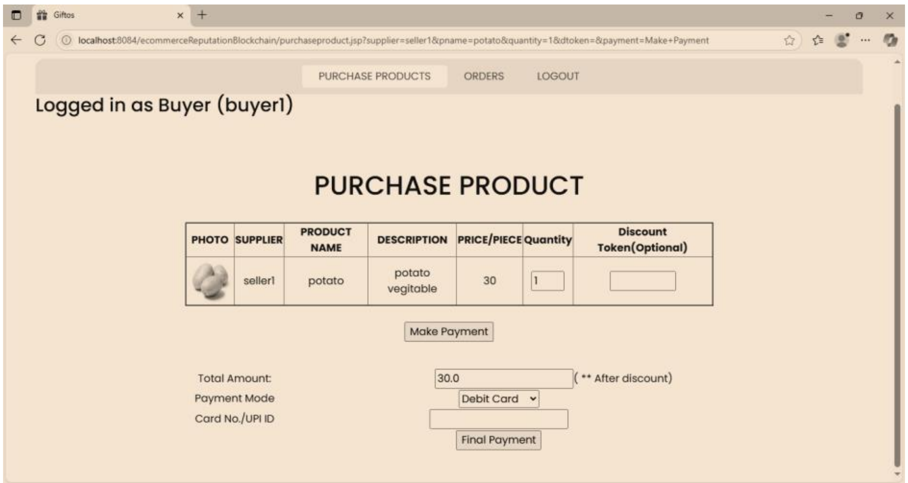

Figure 7.13: The Purchase Product Page (Payment Mode)

The Purchase Product Payment Page allows buyers to complete their transactions by selecting a preferred **payment mode**, such as **Debit Card**, **Credit Card**, or **UPI**. The page displays a table with all relevant product details including **Product Photo**, **Supplier Name**, **Product Name**, **Description, Price**, **Quantity**, and an optional field for entering a **Discount Token**. After confirming the total amount payable, the buyer enters their **Card Number** or **UPI ID** and clicks the payment button to finalize the purchase. This page ensures a smooth, secure, and convenient checkout process within the Trust Chain platform.

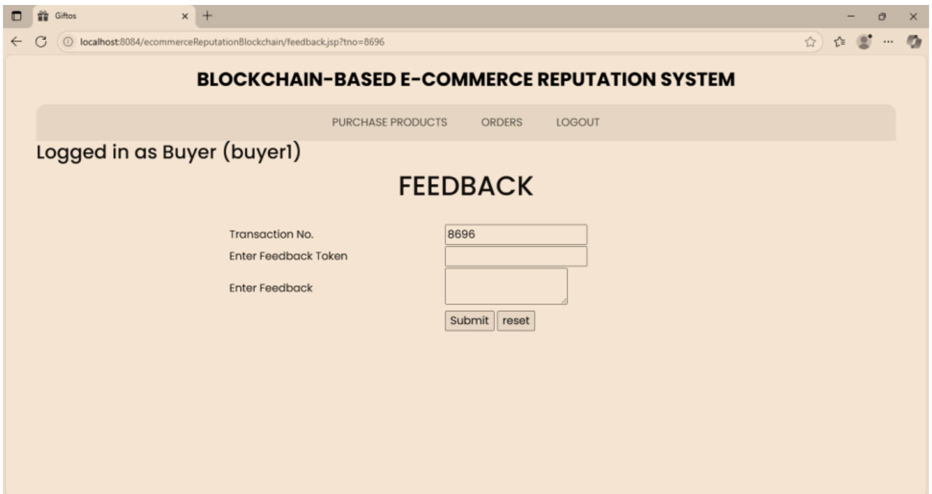

Figure 7.14: The Feedback Page

The Feedback Page allows buyers to provide feedback for products they have purchased. The form includes fields such as **Transaction Number**, **Feedback Token**, and a text area to **Enter Feedback**. After filling in the required information, the buyer can click the Submit button to record their feedback. The system verifies the feedback token with blockchain records to ensure authenticity before accepting the entry. This process helps maintain trust and ensures that only verified buyers can contribute to a seller’s reputation.

CHAPTER 8

CONCLUSION AND FUTURE WORK

CHAPTER 8

CONCLUSION

The exponential growth of e-commerce platforms over the past decade has redefined how buyers and sellers interact, transact, and build trust in digital environments. However, despite the conveniences offered by modern e-commerce systems, **reputation management remains one of its most vulnerable and under-regulated aspects**. Traditional centralized platforms continue to rely on reputation models that are **opaque, manipulable, and biased**, exposing users to risks such as **fraudulent reviews**, **unverified seller identities**, and **reputation inflation or sabotage**. In light of these limitations, this project set out to address the foundational issues of **trust, verification, and transparency** through the design and implementation of an innovative blockchain-based framework—**Trustchain**.

Trustchain is a **privacy-preserving, decentralized reputation system** purpose-built for the e-commerce ecosystem. It leverages the **immutability and transparency** of blockchain technology along with **verifiable credentials** and **smart contracts** to construct a trust architecture where neither users nor administrators can arbitrarily manipulate feedback data. The proposed system replaces traditional, server-controlled rating systems with a model where **feedback is verifiable**, **transactions are traceable**, and **reputation is built on factual, cryptographically validated interactions**.

The system comprises two essential subsystems, each contributing to a broader architecture of reliability and accountability. The **Verifiable Credentials Subsystem** handles the authentication and identity verification of all users, ensuring that only legitimate buyers and sellers participate in the reputation-building process. It uses **digitally signed credentials** stored on a **permissioned blockchain**, enhancing trust without compromising privacy. Sellers are issued cryptographic identities, making it impossible to re-enter the system under false pretenses, while buyers gain a secure and verifiable identity that links them to their transactions.

The second component, the **Feedback Collection Subsystem**, introduces a robust mechanism for **tokenized feedback management**. When a buyer completes a transaction, they are issued a **feedback token**, which acts as a one-time, blockchain-verified key enabling them to submit a review. This prevents review spam, ballot-stuffing, and fake reviews, ensuring that **only genuine buyers can influence a seller's reputation**. Upon submitting valid feedback, buyers are rewarded with **discount tokens**, which incentivize active participation and can be redeemed during future purchases, fostering loyalty while maintaining system integrity.

This architecture is built on **permissioned blockchain platforms such as Hyperledger Fabric and Indy**, selected for their scalability, security, and enterprise-grade consensus protocols. The use of **smart contracts** automates processes like token issuance, feedback validation, and reward generation—removing human bias and enforcing system rules impartially. The blockchain ledger, by design, guarantees **immutability** and **auditability**, allowing for independent verification of any transaction or feedback record without the need for a central authority.

Throughout the development lifecycle, the Trustchain system underwent **rigorous testing across all functional and non-functional dimensions**. From unit-level verification of APIs and modules to full-system integration testing, the platform was evaluated under multiple scenarios to ensure **reliability, security, and performance**. The feedback submission and token verification processes were particularly tested for their **resilience against manipulation**, and the results consistently demonstrated the robustness of blockchain-backed reputation tracking.

In practical terms, Trustchain not only enhances the **credibility and reliability** of e-commerce interactions but also empowers users by returning **ownership and control of reputation data** to the community. Unlike traditional platforms where user data and reviews are locked into proprietary systems, Trustchain ensures **data portability**, **cross-platform reputation interoperability**, and **community-governed transparency**—key principles for the next generation of decentralized digital commerce.

Moreover, the modular design of the system allows for **future extensibility and adaptation**. The architecture supports seamless integration with existing platforms via APIs and can be extended to include **multi-platform token usage**, **cross-border e-commerce compatibility**, and **real-time fraud detection using AI-enhanced scoring models**. Its privacy-preserving mechanisms—especially through **zero-knowledge proofs** and **credential-based selective disclosure**—position Trustchain as a forward-looking solution aligned with global data protection standards such as **GDPR** and **Web 3.0 ethics**.

In conclusion, this project validates the potential of blockchain technology in transforming how trust is established and maintained in digital commerce. Trustchain stands not just as a theoretical model, but as a functional prototype that bridges the gap between **security**, **user trust**, and **systemic fairness**. It opens a new chapter in the evolution of e-commerce, where **trust is no longer issued by**

platforms but earned through cryptographically validated experience, laying a robust foundation for decentralized reputation systems in an increasingly digital world.

Future enhancements

While the current implementation of **Trustchain** successfully demonstrates a functional and secure decentralized reputation system for e-commerce, the scope for further innovation remains significant. The system establishes a strong foundation by leveraging **blockchain technology**, **verifiable credentials**, and **token-based incentives** to foster trust and transparency. However, as with any prototype developed within controlled conditions, transitioning to large-scale real-world deployment introduces several new challenges and opportunities for improvement. This section outlines key areas for **future research, development, and enhancement** that can extend the system's impact, scalability, and usability.

1. Cross-Platform Integration and Interoperability

One of the foremost challenges in bringing Trustchain to real-world adoption is the **diversity of existing e-commerce ecosystems**. Each platform may use distinct architectures, data models, authentication protocols, and business logic. To ensure Trustchain's adoption across multiple platforms, future work must focus on developing **standardized Application Programming Interfaces (APIs)** and **flexible middleware layers** that can act as a bridge between the blockchain-based reputation system and proprietary e-commerce platforms.

This includes building:

- **Platform-agnostic connectors** to interface with different databases and APIs.
- **Adaptable data mappers** to convert native platform schemas into a blockchain-compatible format.
- **Plugin-based modules** to allow for selective integration of specific Trustchain functionalities (e.g., token issuance or feedback validation).

Achieving true interoperability would not only enhance adoption but also enable a **federated reputation ecosystem**, where trust scores and feedback tokens remain valid and verifiable across platforms.

2. Product-Level Feedback and Enhanced Review Granularity

In its current state, Trustchain links buyer feedback primarily to sellers, which is effective for building trust at the vendor level. However, modern buyers often seek **product-specific trustworthiness**, especially in marketplaces with diverse inventories. Future implementations should enhance the feedback model to:

- Link each feedback entry to the **specific product** purchased within a transaction.
- Display aggregated **product ratings** separate from seller-level scores.
- Enable buyers to flag **suspicious products** (e.g., items with unusually low prices or inconsistent reviews).

Such improvements would empower buyers with **more contextual information**, enhance the system's **fraud detection capabilities**, and offer sellers an incentive to maintain quality at the individual product level.

3. Federated Token Utility and Cross-Platform Incentives

Currently, **discount tokens** generated through feedback can be redeemed only within the same platform. To increase the utility and perceived value of these tokens, a future enhancement could involve **federated token acceptance**, wherein multiple e-commerce providers agree to:

- Accept **common discount or reward tokens**.
- Maintain a shared ledger of token transactions through cross-chain protocols.
- Allow **token exchange** or interoperability using technologies such as atomic swaps or token bridges.

This would not only boost **user engagement and feedback participation** but also create an ecosystem of **tokenized incentives** across platform boundaries—an essential step toward building a decentralized e-commerce network.

4. Intelligent and Adaptive Reputation Scoring

The current model calculates reputation in a static, rule-based manner. While effective in controlled environments, more **intelligent and adaptive scoring algorithms** are required to handle the complexity of real-world behavior. Future systems can incorporate:

- **Machine learning models** that analyze historical feedback patterns to detect anomalies or manipulation attempts.
- **Weighted scoring** based on transaction history, feedback consistency, and buyer/seller verification levels.
- Real-time **reputation updates** based on behavioral trends rather than isolated events.

This approach would allow for more **accurate, fraud-resistant, and dynamically adjusted** reputation scores that evolve with user activity and trustworthiness.

5. UI/UX and Accessibility Enhancements

As blockchain-based systems can be complex for non-technical users, future versions of Trustchain should focus heavily on refining the **user interface** and **user experience**. Enhancements may include:

- A **streamlined dashboard** for buyers and sellers to easily view transactions, token history, and reputation scores.
- **Feedback submission wizards** with visual prompts and auto-filled transaction details.
- Accessibility support for **multilingual interfaces** and **mobile devices** to expand user reach across demographics.

A smoother and more intuitive interface will be critical to **mainstream adoption**, especially among users unfamiliar with blockchain concepts.

6. Modular Deployment and Customization

For broader adoption, Trustchain must be deployable in **modular segments**, allowing e-commerce platforms to **selectively adopt** features such as:

- Only verifiable credential management,
- Only tokenized feedback systems,
- Or a full-stack Trustchain deployment.

Such modularity will enable flexible use based on platform needs and regulatory requirements, making the system **adaptable, scalable, and easier to integrate** without requiring a complete infrastructure overhaul.

7. Multi-Blockchain Compatibility and Scalability

Currently designed around **Hyperledger Fabric and Indy**, the system can benefit from expanding its support to **other blockchain platforms** such as:

- **Ethereum** (for DeFi and public token systems),
- **Polygon** (for scalability and low-cost transactions),
- **Solana** (for high-throughput performance),
- Or **Polkadot** (for cross-chain compatibility).

This will ensure the system is not limited by the constraints of a single blockchain and can be deployed in environments with varying performance and trust requirements. Additionally, **cross-chain bridges** or **sidechains** could be implemented to offload high-frequency operations and reduce costs.

Summary of Key Future Enhancements

- **Cross-Platform Integration**: Build APIs and middleware to connect with multiple e-commerce platforms.

- **Product-Level Review Linking**: Attach feedback directly to products as well as sellers for higher review granularity.
- **Federated Token Utility**: Enable the use of discount tokens across multiple platforms within a shared ecosystem.
- **Advanced Reputation Scoring**: Incorporate AI/ML to detect anomalies and improve trust scoring accuracy.
- **UI/UX Enhancements**: Redesign interfaces to improve usability, especially for non-technical users.
- **Modular Deployment**: Allow platforms to adopt Trustchain in parts, based on their infrastructure readiness.
- **Multi-Blockchain Support**: Extend compatibility to public and scalable blockchain platforms for flexibility and reach.

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CONFERENCE PAPER

Trust Chain: Blockchain Powered E-Commerce Reputation System

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***Abstract*— Trust and reputation are critical components in modern e-commerce platforms, where users often rely on feedback and ratings to make informed decisions. However, traditional reputation systems are prone to manipulation, identity fraud, and a lack of transparency due to their centralized architecture. To address these issues, this paper proposes Trust Chain, a blockchain-powered e-commerce reputation system that ensures secure, tamper-proof, and privacy-preserving feedback management. Leveraging Hyperledger Indy for decentralized identity and Hyperledger Fabric for permissioned blockchain infrastructure, the system enables users to submit verifiable reviews without revealing personal information. It incorporates Verifiable Credentials, Zero-Knowledge Proofs (ZKP), and smart contracts to authenticate feedback, automate reputation scoring, and prevent fraudulent activities. Additionally, a token-based incentive mechanism is introduced to encourage honest participation. The proposed solution enhances trust and transparency in e-commerce by providing a decentralized and scalable reputation framework suitable for real-world deployment.**

***Keywords*— Hyperledger Indy, Hyperledger Fabric, Blockchain**

I. INTRODUCTION

This The rapid growth of e-commerce has transformed the global marketplace, enabling consumers and merchants to interact across geographical boundaries with unprecedented ease. A cornerstone of this digital ecosystem is the reputation system, which helps establish trust between anonymous parties through user-generated feedback and ratings. However, existing centralized reputation mechanisms are prone to manipulation, censorship, and lack of transparency. Fake reviews, biased moderation, and identity fraud undermine the reliability of such systems, posing significant risks to users and businesses alike.

Blockchain technology, with its inherent properties of decentralization, immutability, and transparency, offers a promising solution to these challenges. By leveraging blockchain, reputation data can be securely stored and verified without reliance on a single controlling entity. This paper proposes "Trust Chain," a blockchain-powered e-commerce reputation framework that integrates Hyperledger Indy for decentralized identity management and Hyperledger Fabric for secure feedback handling. The system ensures privacy-preserving, tamper-proof, and transparent feedback processing, addressing the shortcomings of conventional reputation models. Through this approach, the proposed system aims to establish a scalable and trustworthy infrastructure for enhancing user confidence and integrity in e-commerce transactions.

II. RELATED WORK

In [1], Sun et al. proposed a blockchain-based feedback token system that allows only verified buyers to submit reviews, mitigating ballot-stuffing attacks. However, the solution was limited by high transaction fees on public blockchains. In a similar direction,

Li et al. [2] developed RepChain, a privacy-preserving reputation model that employs blind signatures to anonymize users. While effective in protecting identities, the model lacked incentive mechanisms for honest feedback.

The DTrust framework by Dhakal et al. [3] introduced a decentralized trust management system using Ethereum

smart contracts. It automated review validation but was constrained by the cost of on-chain data processing and limited scalability. EthReview by Zulfiqar et al. [4] enhanced this approach by integrating smart contracts for secure product reviews. However, it failed to ensure the privacy of reviewers.
Kugblenu and Vuorimaa [5] explored the use of permissioned blockchains for e-commerce reputation management. Their model improved performance and governance but lacked strong cryptographic verification of feedback authenticity. Another relevant work is Beaver [6], a decentralized e-commerce platform that includes a built-in reputation system. Despite its full decentralization, it requires an entirely new marketplace infrastructure, limiting its adaptability to existing platforms.
The proposed system, Trust Chain, builds upon these foundations by integrating Hyperledger Indy for decentralized identity and verifiable credentials, and Hyperledger Fabric for smart contract execution and tamper-proof review storage. It addresses the shortcomings of prior models by offering scalability, privacy, and incentive mechanisms within a permissioned blockchain architecture.

III. METHODLOGY

The proposed system, **Trust Chain**, combines permissioned blockchain architecture with cryptographic verification and smart contracts to build a **transparent and privacy-preserving e-commerce reputation system**. The system leverages **Hyperledger Fabric** for blockchain network implementation and **Hyperledger Indy** for decentralized identity management using **Verifiable Credentials (VCs)** and **Zero-Knowledge Proofs (ZKP)**.

A. Tools and Technologies Used

- **Hyperledger Fabric**: A modular, permissioned blockchain framework used for deploying smart contracts (chain code), managing transaction consensus, and maintaining an immutable ledger of buyer-seller interactions and feedback.
- **Hyperledger Indy**: A distributed ledger purpose-built for decentralized identity management, enabling the generation and verification of Verifiable Credentials.
- **Zero-Knowledge Proofs (ZKP)**: Used to validate credentials without disclosing user identity during the review process.
- **Smart Contracts (Chaincode)**: Developed using Go/JavaScript to automate reputation score calculation and validation logic on the blockchain.
- **Tokenization System**: Two types of tokens are utilized — **Feedback Tokens (FT)** for review submission and **Discount Tokens (DT)** as incentives for verified reviewers.

B. Identity and Credential Verification

1. Upon registration, the **Issuer (trusted authority)** generates a Verifiable Credential for each user (buyer/seller).
2. Users receive **Decentralized Identifiers (DIDs)** linked to their credentials.
3. During feedback submission, users generate a **ZKP presentation** using Hyperledger Indy to prove ownership of the credential without revealing private details.

C. Feedback Verification Algorithm

The feedback verification and reward process consist of the following steps:

1. Buyer makes a purchase and is issued a **Feedback Token (FT)**.
2. Buyer submits feedback along with a ZKP-protected credential.
3. Smart contract (chaincode) on Hyperledger Fabric:
   - Validates the FT.
   - Verifies ZKP authenticity.
   - Stores the feedback immutably in the blockchain.
4. If all checks pass, a **Discount Token (DT)** is generated and issued to the buyer.

D. Reputation Score Calculation Algorithm

The reputation score **R** for a seller is calculated as:

$$R = \sum_{i=1}^{n} (w_i \cdot f_i)$$

Where:

- fi: Feedback rating from buyer *i* (on a normalized scale)
- wi: Weight factor based on buyer's credibility (derived from Verifiable Credentials)
- n: Total number of verified feedback entries

F. System Security & Privacy

The system ensures:

- **Tamper-proof feedback** through immutable ledger storage.
- **Anonymity** via ZKP and decentralized identifiers.
- **Selective disclosure** of attributes through verifiable credentials.
- **Replay attack prevention** using one-time tokens and timestamped signatures.

IV. RESULTS

The proposed system, **Trust Chain**, was successfully implemented using **Hyperledger Indy** and **Hyperledger Fabric** to evaluate its effectiveness in securing and managing e-commerce reputation. The integration of verifiable credentials with Zero-Knowledge Proofs (ZKP) ensured that only legitimate, privacy-preserving reviews were accepted. Feedback and reputation data were stored immutably on the blockchain, ensuring tamper-proof transparency.

Performance testing demonstrated the system's ability to:

- **Authenticate users** securely via decentralized identity credentials.
- **Prevent fake reviews** using feedback tokens and smart contract validation.
- **Maintain privacy** through the use of ZKP, ensuring reviewers' anonymity.
- **Efficiently calculate reputation scores** using weighted algorithms embedded in Fabric chaincode.

Compared to traditional centralized reputation systems, the blockchain-based approach showed significant improvement in terms of **trustworthiness, resilience to manipulation, and auditability**. The feedback submission process remained fast and scalable, thanks to the modular design of the permissioned blockchain infrastructure.

V. CONCLUSION

This paper presented **Trust Chain**, a blockchain-powered reputation system for e-commerce platforms, addressing the key limitations of conventional feedback models. By combining **Hyperledger Indy** for decentralized identity management and **Hyperledger Fabric** for tamper-resistant feedback storage and smart contract execution, the system achieved **privacy preservation**, **security**, and **transparency**.

The inclusion of **feedback and discount tokens** ensured user accountability and incentivized honest participation. Additionally, the implementation of **Zero-Knowledge Proofs** allowed users to validate their credentials without compromising anonymity. This approach provides a scalable and efficient foundation for next-generation e-commerce platforms to build reliable trust frameworks, laying the groundwork for broader adoption in decentralized marketplaces.

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Department of Computer Science & Engineering

For participating and presenting a paper titled

"Trust Chain: Blockchain Powered E Commerce Reputation System"

In International Conference on *Global Convergence in Technology, Entrepreneurship, Computing & Value Engineering: Principles and Practices, ICGCP-2025* organized by Sapthagiri College of Engineering, Bengaluru during 16–18th April, 2025

Dr. R.G.Deshpande
Conference Co-ordinator

Dr.Tulsidas .D
Conference -Chair
Principal -SCE

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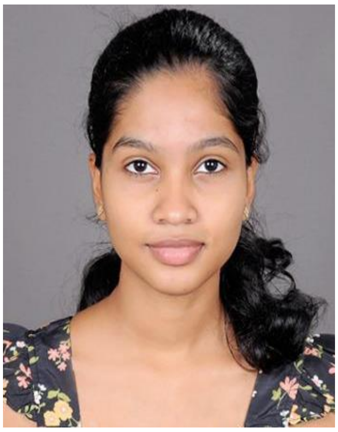
<td>
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